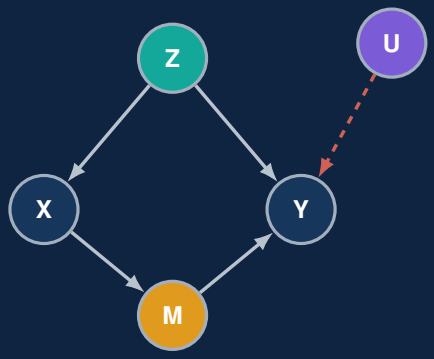


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 13

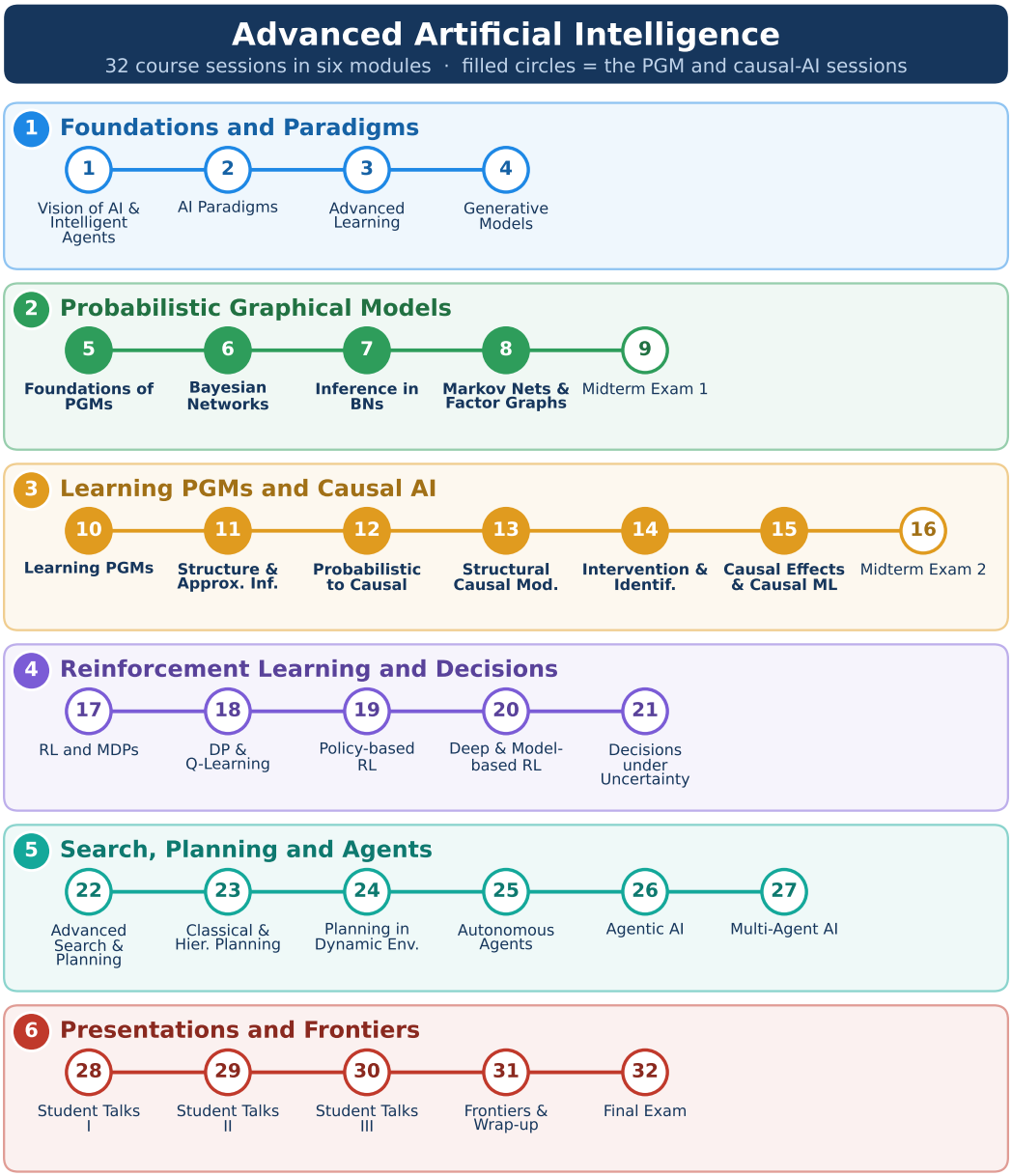
Structural Causal Models

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 13 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, a "you are here" course map, the core theory organized in parts, a dedicated part on mathematical statistics, fully worked numerical and analytical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s13_examples.py` for the numbers, `python/s13_figures.py` for the figures), and `questions/s13_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions.

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SESSION 13 ADVANCED ARTIFICIAL INTELLIGENCE

Structural Causal Models

Structural equations, causal DAGs, the causal Markov condition, d-separation and counterfactuals

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Structural Equations
Mechanisms, assignments, push-forward; structural vs statistical equations.

PART II
Causal DAGs & d-Separation
Causal Markov condition, testable implications, faithfulness, latent confounders.

PART III
Interventions & Counterfactuals
Graph surgery, truncated factorization, abduction–action–prediction, PN/PS.

PART IV
Mathematical Statistics
Linear SEMs, path tracing, omitted-variable bias, LiNGAM, Gaussian abduction.

PART V
AI Connections
World models, agent what-ifs, COMA credit assignment, counterfactual XAI.

LEARNING OBJECTIVES

- Define a structural causal model $M = (\mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}))$ and explain why $:=$ is an *assignment*, not an equation.
- State and prove the **causal Markov condition**: a Markovian SCM induces a distribution that factorizes over its causal DAG.
- Read **testable implications** off a causal DAG by d-separation, and recognise faithfulness, minimality and latent confounding.
- Compute interventional distributions by graph surgery and the truncated factorization, and counterfactuals by abduction, action and prediction.
- Derive the covariance of a linear SEM, $\Sigma = (I - B)^{-1}\Omega(I - B)^{-\top}$, apply Wright's path-tracing (trek) rule, and separate regression coefficients from structural coefficients.
- Use SCM reasoning in AI: world models in model-based RL, "what-if" simulation in LLM agents, counterfactual credit assignment (COMA) in multi-agent RL.

Where we are. Sessions 5–11 (course sessions 5–11) built the *probabilistic* machinery: conditional independence, Bayesian and Markov networks, exact and approximate inference, parameter and structure learning. Session 12 made the conceptual pivot: correlation is not causation, $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$, and confounding and selection bias explain the gap. Today we build the mathematical object that makes $\mathrm{do}(\cdot)$ and "what would have happened" well defined: the **structural causal model**. Session 14 will then ask when causal quantities can be *identified* from observational data,

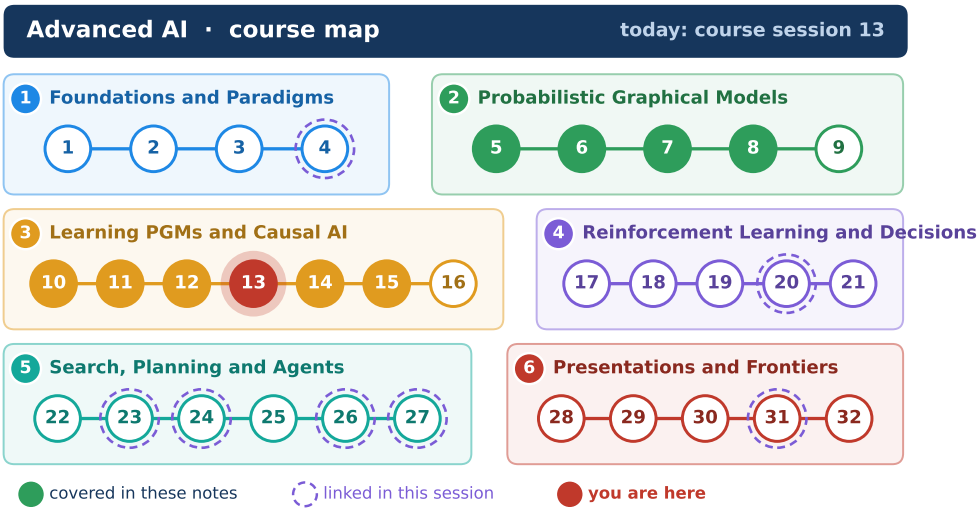


Figure 13.1. Where this session sits in the Advanced AI course. Dashed purple rings mark the course sessions we link to today.

and Session 15 will estimate them with machine learning. Beyond the notes, SCMs reappear in model-based RL, planning, agentic and multi-agent AI (course sessions 20–27, 31).

OPENING MOTIVATION

Session 12 ended on a gap: $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$ in general, and nothing in our toolkit so far can compute the right-hand side, because a DAG with CPTs only ever specified a *factorization* of P — it never said what happens if a piece of that factorization is overridden rather than observed. Today we fix this by strengthening what an arrow means. Instead of "X_i's distribution depends on its parents," a structural causal model (SCM) says "X_i is *computed from* its parents by a fixed mechanism" — a claim strong enough to survive having one mechanism swapped out, which is exactly what an intervention does.

Healthcare

"Would *this* patient have recovered without the drug?" is a rung-3 counterfactual: it needs mechanisms, not just a joint distribution.

Robotics & RL

A model-based agent learns $s_{t+1} := f(s_t, a_t, u_t)$; planning intervenes on it, replaying a logged episode is a counterfactual.

LLM agents

An agent simulates $P(\text{success} \mid \mathrm{do}(\text{tool}))$ before acting, and afterwards asks whether the call was *necessary*.

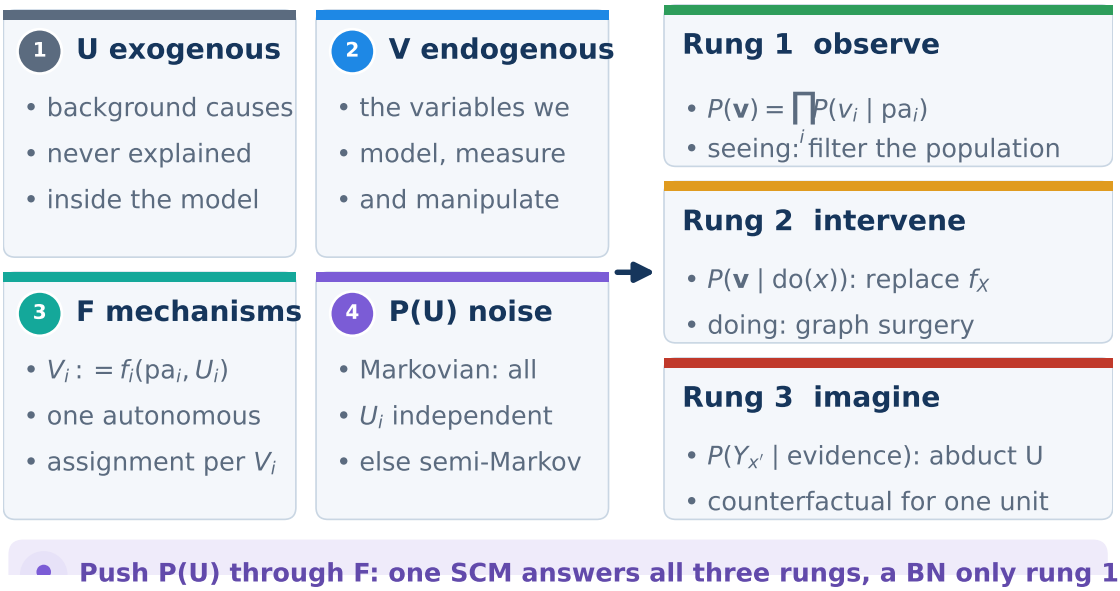


Figure 13.2. Anatomy of an SCM. The four ingredients (left) are pushed forward into the three kinds of query of Pearl's ladder (right): observational, interventional and counterfactual.

PART I

Structural Equations

Mechanisms, not merely factorizations

PART I · STRUCTURAL EQUATIONS

13.1 Structural Causal Models

Definition 13.1 (Structural causal model)**.** A structural causal model $M = (\mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}))$ consists of:

- *exogenous* variables $\mathbf{U}$ — background factors external to the model, with a specified joint distribution $P(\mathbf{U})$;
- *endogenous* variables $\mathbf{V} = \{V_1, \dots, V_n\}$ — the variables of interest, determined within the model;
- a set of *structural equations* $\mathcal{F} = \{f_1, \dots, f_n\}$, one per endogenous variable, of the form

$$V_i := f_i\big(\mathrm{pa}(V_i), U_i\big),$$

where $\mathrm{pa}(V_i) \subseteq \mathbf{V} \setminus \{V_i\}$ and $U_i \subseteq \mathbf{U}$ are the direct causes of V_i.

The symbol $:=$ (assignment, not equality) is deliberate: $V_i := f_i(\cdots)$ says V_i is *computed* by this mechanism, not merely that this equation happens to hold. This asymmetry — present in the notation but invisible in an ordinary equals sign — is precisely what will let us "delete" one equation while keeping the rest, which an algebraic equality could never justify (you cannot selectively erase

one equation from a system of equalities without changing what the whole system claims).

Every SCM induces an ordinary graph G: draw an edge $V_j \to V_i$ whenever $V_j \in \mathrm{pa}(V_i)$. We call M *recursive* (acyclic) if G is a DAG, and *Markovian* if in addition the noises $U_1, \ldots, U_n$ are mutually independent. Unless stated otherwise all SCMs in these notes are recursive. And crucially:

Proposition 13.1 (Consistency with Bayesian networks (Session 6))**.** *The distribution $P(\mathbf{v})$ induced by a Markovian SCM (obtained by pushing $P(\mathbf{U})$ forward through $\mathcal{F}$) factorizes exactly as $\prod_i P(v_i \mid \mathrm{pa}(v_i))$ over the induced DAG G — the same Bayesian network factorization from Session 6.*

So nothing from Sessions 5–11 is discarded: an SCM is a *refinement* of a Bayesian network, adding mechanism on top of factorization, and every observational computation (VE, MLE, EM, sampling) still applies unchanged to the induced $P(\mathbf{v})$. What's new is that the SCM also licenses questions the factorization alone could not answer. Proposition 13.1 is proved in full as the causal Markov condition, Theorem 13.1 below.

Worked Example 13.1 (A linear SCM for wages)**.** Let A (ability), X (years of education), Y (income) be linked by

$$A := U_A, \qquad X := 2A + U_X, \qquad Y := 3X + 5A + U_Y,$$

with U_A, U_X, U_Y mutually independent standard normal noises (Fig. 13.3). For later use we name the three structural coefficients $a = 2$ (edge $A \to X$), $b = 3$ (edge $X \to Y$) and $c = 5$ (edge $A \to Y$). This is exactly the confounding structure of Session 12: A is a common cause of X and Y (a fork), so observing X carries information about A, which in turn carries information about Y through the direct arrow $A \to Y$ — inflating any naive association between X and Y beyond the true structural coefficient 3.

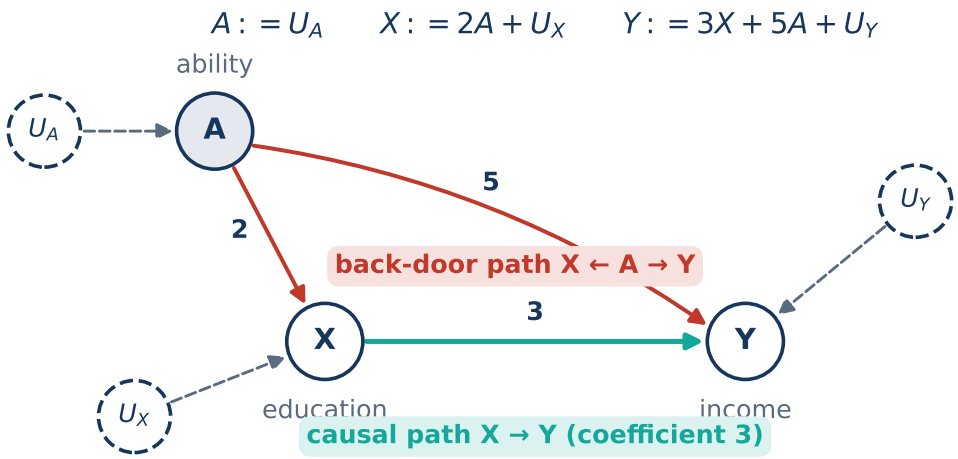


Figure 13.3. An SCM as a generative mechanism: solid nodes are endogenous, dashed nodes are exogenous noise; edge labels are structural coefficients. The back-door path $X \leftarrow A \to Y$ is the source of confounding.

PART I · STRUCTURAL EQUATIONS

13.2 Structural versus Statistical Equations

A regression equation and a structural equation can look identical on paper. The difference is what they promise under change.

Worked Example 13.2 (One mechanism, two regressions)**.** Let $X := U_X$, $Y := 2X + U_Y$ with $U_X, U_Y \overset{\text{iid}}{\sim} N(0,1)$. Then (X, Y) is bivariate normal with zero means and

$$\mathrm{Var}(X) = 1, \qquad \mathrm{Var}(Y) = 2^2\mathrm{Var}(X) + \mathrm{Var}(U_Y) = 4 + 1 = 5,$$

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, 2X + U_Y) = 2\mathrm{Var}(X) = 2.$$

The best linear predictor of X from Y has slope $\mathrm{Cov}(X,Y)/\mathrm{Var}(Y) = 2/5 = 0.4$, i.e. $X = 0.4\,Y + E$, where $E = X - 0.4Y$ is uncorrelated with Y (hence, being jointly Gaussian with Y, independent of it) and $\mathrm{Var}(E) = \mathrm{Var}(X) - \mathrm{Cov}(X,Y)^2/\mathrm{Var}(Y) = 1 - 4/5 = 0.2$: a perfectly valid *statistical* equation. But it is not *structural*. Under $\mathrm{do}(Y = 5)$ the SCM replaces only Y's assignment; $X := U_X$ is untouched, so $\mathbb{E}[X \mid \mathrm{do}(Y = 5)] = \mathbb{E}[U_X] = 0$, whereas the regression "predicts" $\mathbb{E}[X \mid Y = 5] = 0.4 \cdot 5 = 2$. And solving $Y := 2X + U_Y$ for $X = (Y - U_Y)/2$ is algebraically valid but causally meaningless: it would claim that setting Y moves X.

PYTHON LAB · `s13_examples.py` · `ex_structural_vs_statistical()`

Simulation of the SCM with $n = 100\,000$ draws (`default_rng(11)`); the interventional mean uses fresh draws of $X := U_X$.

```
exact:  slope Cov/Var(Y) = 2/5 = 0.40, residual var = 0.20
simulated slope = 0.400, residual var = 0.199, corr(resid, Y) = -0.0000
E[X | do(Y=5)] (simulated) = -0.001  vs  E[X | Y=5] = 0.4*5 = 2.0
```

Table 13.1. Structural versus statistical equations.

	Statistical (regression) equation	Structural equation
Symbol	$Y = \beta X + \varepsilon$	$Y := f_Y(X, U_Y)$
Claim	best predictor of Y from X in *this* population	mechanism that *generates* Y from its causes
Error term	orthogonal to X by construction	independent of non-descendants by assumption (Markovian)
Under $\mathrm{do}(X = x)$	no promise	still valid (autonomy / invariance)

The property in the last row of Table 13.1 has a name: **autonomy** or **modularity** — each mechanism f_i can be changed without changing the others. It is the assumption that makes interventions predictable, and it is what a world model in RL or a simulator inside a planning agent silently relies on.

Worked Example 13.3 (Two SCMs, one distribution, two causal answers)**.** Let M_1: $X := U_X$, $Y := 0.6X + U_Y$ with $U_X \sim N(0,1)$, $U_Y \sim N(0, 0.64)$; and M_2: $Y := V_Y$, $X := 0.6Y + V_X$ with $V_Y \sim N(0,1)$, $V_X \sim N(0, 0.64)$. Both induce a zero-mean bivariate normal law. Under M_1: $\mathrm{Var}(X) = 1$, $\mathrm{Var}(Y) = 0.6^2 \cdot 1 + 0.64 = 0.36 + 0.64 = 1$ and $\mathrm{Cov}(X, Y) = 0.6\mathrm{Var}(X) = 0.6$; under

M_2, by the same computation with the roles swapped, $\mathrm{Var}(Y) = 1$, $\mathrm{Var}(X) = 0.36 + 0.64 = 1$ and $\mathrm{Cov}(X, Y) = 0.6$. A zero-mean Gaussian law is determined by its covariance matrix, so both models induce the same law (unit variances, correlation 0.6), and no amount of observational data can distinguish them. Yet under M_1 surgery keeps $Y := 0.6X + U_Y$, so $\mathbb{E}_{M_1}[Y \mid \mathrm{do}(X = 2)] = 0.6 \cdot 2 = 1.2$, while under M_2 the variable Y is a root whose assignment $Y := V_Y$ is untouched by $\mathrm{do}(X = 2)$, so $\mathbb{E}_{M_2}[Y \mid \mathrm{do}(X = 2)] = \mathbb{E}[V_Y] = 0$. The same construction (regress either variable on the other, Gaussian residual) works for *every* bivariate normal law. Observational equivalence does not imply interventional equivalence; this is the SCM version of Session 6's Markov equivalence, and the reason Session 14 needs identification theory.

PYTHON LAB · s13_examples.py · ex_markov_equivalent_gauss()

```
M1:  Var(Y) = 0.6^2*1 + 0.64 = 1.00, Cov(X,Y) = 0.6;  M2:  Var(X) = 1.00,
Cov = 0.6
Cov under M1 = Cov under M2 = [[1.  0.6] [0.6 1.  ]]
E[Y | do(X=2)]:   M1 -> 0.6*2 = 1.2,   M2 -> E[V_Y] = 0.0
```

Table 13.2. The three rungs of causal queries answered by one SCM.

	Observational	Interventional	Counterfactual
Query	$P(y \mid x)$	$P(y \mid \mathrm{do}(x))$	$P(Y_{x'} = y' \mid X = x, Y = y)$
Operation	condition on $X = x$	replace f_X by x (surgery)	abduct $\mathbf{U}$, then surgery
Graph used	G	$G_{\overline{X}}$ (no edges into X)	$G_{\overline{X}}$ with posterior noise
Needs	$P(\mathbf{v})$	$P(\mathbf{v})$ + graph (if identifiable)	full $\mathcal{F}$ and $P(\mathbf{U})$ in general
Example	"Do educated people earn more?"	"What if we mandate 2 more years?"	"Would *she* earn more with 2 more years?"

PART II

Causal DAGs, the Causal Markov Condition and d-Separation

What a causal graph claims, and how the data can refute it

PART II · CAUSAL MARKOV

13.3 Causal DAGs and the Causal Markov Condition

Definition 13.2 (Causal DAG)**.** A DAG G over $\mathbf{V}$ is a *causal DAG* for a system if (i) every edge $V_j \to V_i$ means V_j is a *direct cause* of V_i relative to $\mathbf{V}$ (setting V_j by intervention can change V_i even when all other variables in $\mathbf{V}$ are held fixed), and (ii) every common cause of two or more variables of $\mathbf{V}$ is itself in $\mathbf{V}$ (*causal sufficiency*). Equivalently, G is the graph of a

Markovian SCM.

The *presence* of an edge is a weak claim (the effect may be tiny); the *absence* of an edge is a strong one (no direct effect) and, with causal sufficiency, generates the testable implications below.

Theorem 13.1 (Causal Markov condition)**.** *Let M be a Markovian SCM with causal DAG G. Then for every i,*

$$V_i \perp\!\!\!\perp \mathrm{nd}(V_i) \setminus \mathrm{pa}(V_i) \mid \mathrm{pa}(V_i) \qquad \textit{(local Markov property)},$$

and consequently the induced distribution factorizes as $P(\mathbf{v}) = \prod_{i=1}^{n} P(v_i \mid \mathrm{pa}(v_i))$, with $P(v_i \mid \mathrm{pa}_i) = P\big(f_i(\mathrm{pa}_i, U_i) = v_i\big)$.

Proof. Order the variables topologically. By induction along this order, each V_j is a (measurable) function of the noises $\{U_k : k \in \mathrm{an}(j) \cup \{j\}\}$ of itself and its ancestors. Fix i and let $N = \mathrm{nd}(V_i)$. If $V_j \in N$, then neither V_i nor any descendant of V_i is an ancestor of V_j (otherwise V_j would be a descendant of V_i). Hence $N = g(\mathbf{U}_S)$ for an index set S with $i \notin S$, and Markovianity gives $U_i \perp\!\!\!\perp N$. Since $\mathrm{pa}(V_i) \subseteq N$, for any measurable set B,

$$P(V_i \in B \mid N = n) = P\big(f_i(\mathrm{pa}_i(n), U_i) \in B \mid N = n\big) = P\big(f_i(\mathrm{pa}_i(n), U_i) \in B\big),$$

where the last step is the "freezing" lemma: if $U \perp\!\!\!\perp N$ then $P(h(N,U) \in B \mid N = n) = P(h(n,U) \in B)$, by Fubini's theorem on the product law $P_N \otimes P_U$. The right-hand side depends on n only through $\mathrm{pa}_i(n)$, which is the local Markov property. For the factorization, apply the chain rule along the topological order: the predecessors $V_{<i}$ are non-descendants of V_i, so $P(v_i \mid v_{<i}) = P(v_i \mid \mathrm{pa}_i)$. □

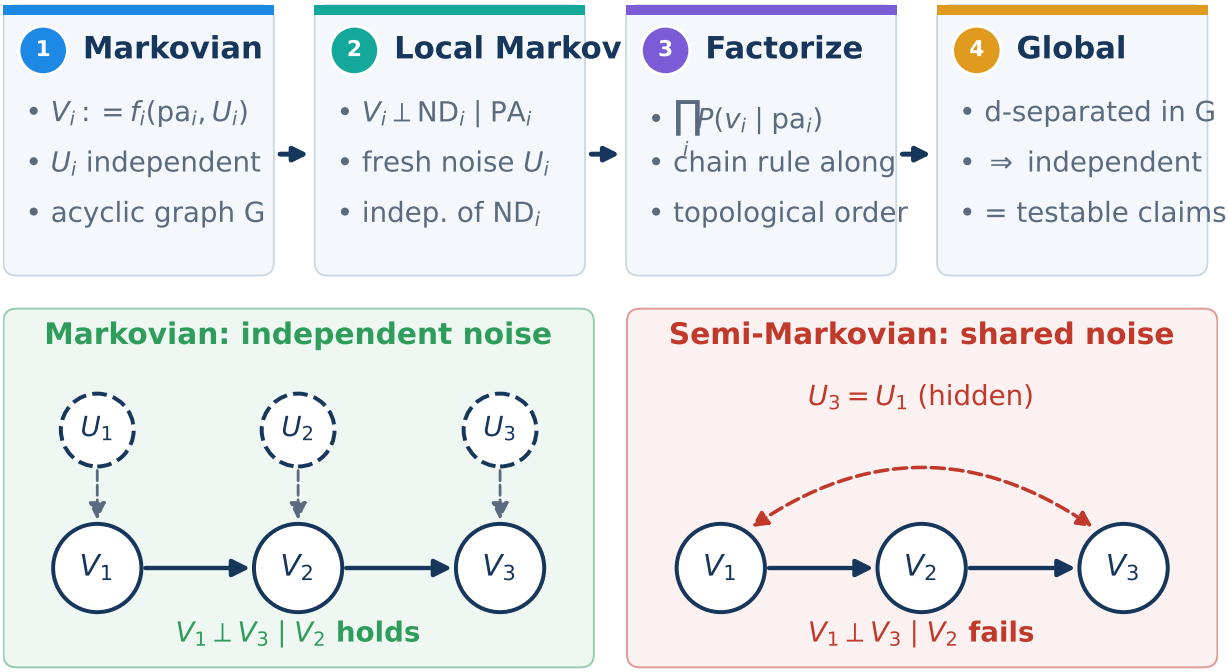


Figure 13.4. The causal Markov condition as a chain of implications (top), and why independent noise matters (bottom): if a hidden factor feeds both U_1 and U_3, the visible DAG no longer implies $V_1 \perp\!\!\!\perp V_3 \mid V_2$.

Worked Example 13.4 (Pushing noise through XOR mechanisms)**.** Let $V_1 := U_1$, $V_2 := V_1 \oplus U_2$, $V_3 := V_2 \oplus U_3$ ($\oplus$ = exclusive or) with independent $U_1 \sim \text{Bern}(0.4)$, $U_2 \sim \text{Bern}(0.1)$, $U_3 \sim \text{Bern}(0.2)$. The map $(u_1, u_2, u_3) \mapsto (v_1, v_2, v_3)$ is one-to-one ($u_1 = v_1$, $u_2 = v_1 \oplus v_2$, $u_3 = v_2 \oplus v_3$), so each of the 8 joint values has the probability of one noise configuration; e.g. $(V_1, V_2, V_3) = (0, 0, 0)$ needs $(U_1, U_2, U_3) = (0, 0, 0)$, so $P(0, 0, 0) = 0.6 \cdot 0.9 \cdot 0.8 = 0.432$.

- *Marginal dependence.* Since $V_3 = V_1 \oplus U_2 \oplus U_3$, given $V_1 = 1$ we have $V_3 = 1$ iff $U_2 \oplus U_3 = 0$: $P(V_3{=}1 \mid V_1{=}1) = 0.9 \cdot 0.8 + 0.1 \cdot 0.2 = 0.72 + 0.02 = 0.74$. Given $V_1 = 0$, $V_3 = 1$ iff $U_2 \oplus U_3 = 1$: $P(V_3{=}1 \mid V_1{=}0) = 0.9 \cdot 0.2 + 0.1 \cdot 0.8 = 0.18 + 0.08 = 0.26$. So V_1 and V_3 are dependent.
- *Conditional independence.* Given $V_2 = v_2$, $V_3 = 1$ iff $U_3 = 1 - v_2$, and U_3 is independent of (U_1, U_2), hence of (V_1, V_2). So $P(V_3{=}1 \mid V_2{=}0, V_1) = P(U_3{=}1) = 0.2$ and $P(V_3{=}1 \mid V_2{=}1, V_1) = P(U_3{=}0) = 0.8$ for both values of V_1: exactly $V_1 \perp\!\!\!\perp V_3 \mid V_2$, and the joint equals $P(v_1)P(v_2 \mid v_1)P(v_3 \mid v_2)$ (checked for all 8 cells in the script).
- *Shared noise.* Now let U_3 *be* U_1 (a shared hidden cause). Then $V_3 = V_2 \oplus U_1 = V_2 \oplus V_1$ is a deterministic function of (V_1, V_2): given $V_2 = 0$, $V_3 = V_1$, so $P(V_3{=}1 \mid V_2{=}0, V_1{=}1) = 1$ but $P(V_3{=}1 \mid V_2{=}0, V_1{=}0) = 0$, and the CI implied by the visible chain fails.

PYTHON LAB · s13_examples.py · ex_causal_markov_discrete()

```
P(V1,V2,V3 = (0, 0, 0)) = 0.4320
P(V3=1|V1=1) = 0.7400, P(V3=1|V1=0) = 0.2600  (dependent)
Markovian:  P(V3=1|V2=1,V1=1) = 0.8000, P(V3=1|V2=1,V1=0) = 0.8000
max |P(v) - P(v1)P(v2|v1)P(v3|v2)| = 6.94e-18
shared noise U3=U1:  P(V3=1|V2=0,V1=1) = 1.0000, P(V3=1|V2=0,V1=0) = 0.0000
```

PART II · CAUSAL MARKOV

13.4 d-Separation in SCMs: Testable Implications

Because the induced distribution factorizes over G, everything we proved about Bayesian networks in Session 6 applies verbatim.

Theorem 13.2 (Global Markov property of SCMs)**.** *If M is a Markovian SCM with causal DAG G, then for disjoint sets $\mathbf{X}, \mathbf{Y}, \mathbf{Z} \subseteq \mathbf{V}$,*

$$\mathbf{X} \perp\!\!\!\perp_d \mathbf{Y} \mid \mathbf{Z} \text{ in } G \quad \Longrightarrow \quad \mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z} \text{ in } P_M.$$

Proof sketch. P_M factorizes over G (Theorem 13.1), and by the soundness theorem of d-separation (Session 6; Verma and Pearl) every such distribution satisfies all d-separations of G. □

Every d-separation in the causal DAG is therefore a **testable implication** of the causal model: a constraint on observational data that the model must satisfy, whatever the mechanisms f_i are. If the data violate it, the causal model (or the Markov assumption) is wrong.

Worked Example 13.5 (Collider partial correlation in closed form)**.** Consider three linear-Gaussian SCMs on (X, Y, Z), all noises independent $N(0, 1)$: the chain $X := U_X$, $Y :=$

$0.8X + U_Y$, $Z := 0.8Y + U_Z$; the fork $Y := U_Y$, $X := 0.8Y + U_X$, $Z := 0.8Y + U_Z$; and the collider $X := U_X$, $Z := U_Z$, $Y := 0.8X + 0.8Z + U_Y$. We use the Gaussian conditioning formulas $\mathrm{Cov}(X, Z \mid Y) = \mathrm{Cov}(X, Z) - \mathrm{Cov}(X, Y)\mathrm{Cov}(Z, Y)/\mathrm{Var}(Y)$ and $\mathrm{Var}(X \mid Y) = \mathrm{Var}(X) - \mathrm{Cov}(X, Y)^2/\mathrm{Var}(Y)$.

- *Collider.* $\mathrm{Var}(X) = \mathrm{Var}(Z) = 1$, $\mathrm{Var}(Y) = 0.64 + 0.64 + 1 = 2.28$, $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Z, Y) = 0.8$, $\mathrm{Cov}(X, Z) = 0$ (so $\rho_{XZ} = 0$). Then $\mathrm{Cov}(X, Z \mid Y) = 0 - 0.64/2.28 = -0.2807$ and $\mathrm{Var}(X \mid Y) = \mathrm{Var}(Z \mid Y) = 1 - 0.64/2.28 = 0.7193$, hence

$$\rho_{XZ \cdot Y} = \frac{-0.64/2.28}{1 - 0.64/2.28} = \frac{-0.64}{2.28 - 0.64} = -\frac{0.64}{1.64} = -0.390.$$

- *Chain.* $\mathrm{Var}(X) = 1$, $\mathrm{Var}(Y) = 0.64 + 1 = 1.64$, $\mathrm{Var}(Z) = 0.64 \cdot 1.64 + 1 = 2.0496$, $\mathrm{Cov}(X, Z) = 0.8 \cdot 0.8 \cdot \mathrm{Var}(X) = 0.64$, so $\rho_{XZ} = 0.64/\sqrt{1 \cdot 2.0496} = 0.64/1.4316 = 0.447$; given Y, $Z - 0.8Y = U_Z$ is independent of (X, Y), so $\rho_{XZ \cdot Y} = 0$.
- *Fork.* $\mathrm{Var}(X) = \mathrm{Var}(Z) = 0.64 + 1 = 1.64$, $\mathrm{Cov}(X, Z) = 0.8 \cdot 0.8 \cdot \mathrm{Var}(Y) = 0.64$, so $\rho_{XZ} = 0.64/1.64 = 0.390$; given Y, the residuals U_X, U_Z are independent, so $\rho_{XZ \cdot Y} = 0$.

Conditioning on a collider *creates* the dependence that conditioning on a chain or fork *removes* (Session 12's Berkson effect).

PYTHON LAB · `s13_examples.py` · `ex_testable_implications()`

Simulation: $n = 5000$ draws from each of the three SCMs (`simulate_triples()`, `default_rng(6)`); partial correlations by the residual method; two-sided Fisher-z tests.

```
chain   :  corr(X,Z) = +0.447 (p < 0.001);  pcorr(X,Z|Y) = -0.008 (p =
0.569)
fork    :  corr(X,Z) = +0.396 (p < 0.001);  pcorr(X,Z|Y) = +0.013 (p =
0.358)
collider:  corr(X,Z) = +0.002 (p = 0.887);  pcorr(X,Z|Y) = -0.395 (p <
0.001)
collider exact:  Var(Y) = 2.28, 0.64/Var(Y) = 0.2807, Var(X|Y) = 0.7193,
pcorr(X,Z|Y) = -0.390
```

(Exact values 0.447, 0.390, −0.390; every d-separation — $X \perp\!\!\!\perp_d Z \mid Y$ in the chain and fork, $X \perp\!\!\!\perp_d Z$ in the collider — shows up as a correlation indistinguishable from zero.)

13.4.1 Faithfulness and minimality

Theorem 13.2 is one-directional: d-separation implies independence, but a distribution may contain *extra* independences.

Definition 13.3 (Faithfulness and minimality)**.** P is *faithful* to G if every CI in P is implied by d-separation in G, i.e. $\mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z} \Rightarrow \mathbf{X} \perp\!\!\!\perp_d \mathbf{Y} \mid \mathbf{Z}$. P satisfies *causal minimality* with respect to G if it is Markov to G but not to any proper subgraph of G (every edge is "used").

Worked Example 13.6 (Cancelling paths)**.** Let $X := U_X$, $M := X + U_M$, $Y := bX + M + U_Y$ with unit-variance independent Gaussian noises. Then $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, bX + M + U_Y) = b\mathrm{Var}(X) + \mathrm{Cov}(X, M) = b \cdot 1 + 1 = b + 1$, so at $b = -1$ the direct and mediated effects cancel exactly: $X \perp\!\!\!\perp Y$ although $X \to Y$ is an edge, and the total effect $\frac{d}{dx}\mathbb{E}[Y \mid \mathrm{do}(x)] = b + 1 = 0$.

The distribution is Markov and minimal but *not faithful*. The violation is visible in other statistics. With $\mathrm{Var}(X) = 1$, $\mathrm{Var}(M) = 1 + 1 = 2$ and $\mathrm{Cov}(X, M) = 1$, Gaussian conditioning gives $\mathrm{Var}(X \mid M) = 1 - 1^2/2 = \frac{1}{2}$; since $Y = bX + M + U_Y$ and M is fixed, $\mathrm{Cov}(X, Y \mid M) = b\mathrm{Var}(X \mid M) = -\frac{1}{2}$ and $\mathrm{Var}(Y \mid M) = b^2\mathrm{Var}(X \mid M) + \mathrm{Var}(U_Y) = \frac{1}{2} + 1 = \frac{3}{2}$, so

$$\rho_{XY \cdot M} = \frac{-1/2}{\sqrt{(1/2)(3/2)}} = \frac{-0.5}{\sqrt{0.75}} = -0.577 \neq 0.$$

The pattern "$X \perp\!\!\!\perp Y$ but $X \not\perp\!\!\!\perp Y \mid M$" is exactly what a *faithful collider* $X \to M \leftarrow Y$ would produce, so a constraint-based learner that assumes faithfulness (such as PC) would output that wrong graph.

PYTHON LAB · `s13_examples.py` · `ex_faithfulness()`

```
Cov(X,Y) exact = b + 1 = -1 + 1 = 0; simulated (n=100000, seed=5) corr(X,Y)
= +0.0020
pcorr(X,Y|M) simulated = -0.5786;  exact via conditional covariances:
-0.5774
```

For linear-Gaussian SCMs with coefficients drawn from a density, unfaithful parameters have Lebesgue measure zero (Spirtes, Glymour and Scheines 2000): faithfulness is a mild *generic* assumption — the one the PC algorithm of Session 11 needs — but near-cancellations (feedback, agents that *compensate* for disturbances) make it fragile in finite samples.

PART II · CAUSAL MARKOV

13.5 Semi-Markovian Models and Bidirected Edges

If two noises are dependent — typically because an unmeasured variable L affects both V_i and V_j — the SCM is *semi-Markovian*. We then drop L from the picture and draw a dashed bidirected edge $V_i \leftrightarrow V_j$, obtaining an *acyclic directed mixed graph* (ADMG). The visible DAG part no longer satisfies Theorem 13.1 (bottom of Fig. 13.4), but d-separation extends to ADMGs (m-separation), treating $V_i \leftrightarrow V_j$ as $V_i \leftarrow L \to V_j$. The wage SCM with A unmeasured is the "bow-arc" ADMG $X \to Y$, $X \leftrightarrow Y$, which Session 14 shows is *not* identifiable; bidirected edges partition $\mathbf{V}$ into *c-components*, the building blocks of the ID algorithm.

PART III

Interventions and Counterfactuals

From seeing, to doing, to imagining

PART III · INTERVENTIONS

13.6 Intervention as Graph Surgery

Definition 13.4 (Intervention)**.** The intervention $\mathrm{do}(X = x)$ on SCM M produces a new (*mutilated*, or *submodel*) SCM M_x by replacing the structural equation for X with the constant assignment $X := x$, deleting X's dependence on $\mathrm{pa}(X)$ and U_X, and leaving every other equation in $\mathcal{F}$ unchanged. The interventional distribution $P(\mathbf{y} \mid \mathrm{do}(x))$ is the distribution of $\mathbf{Y}$ induced by M_x.

Graphically (Fig. 13.5), this deletes every incoming edge to X while keeping all outgoing edges intact — X still causes whatever it caused before, it is simply no longer caused by anything itself.

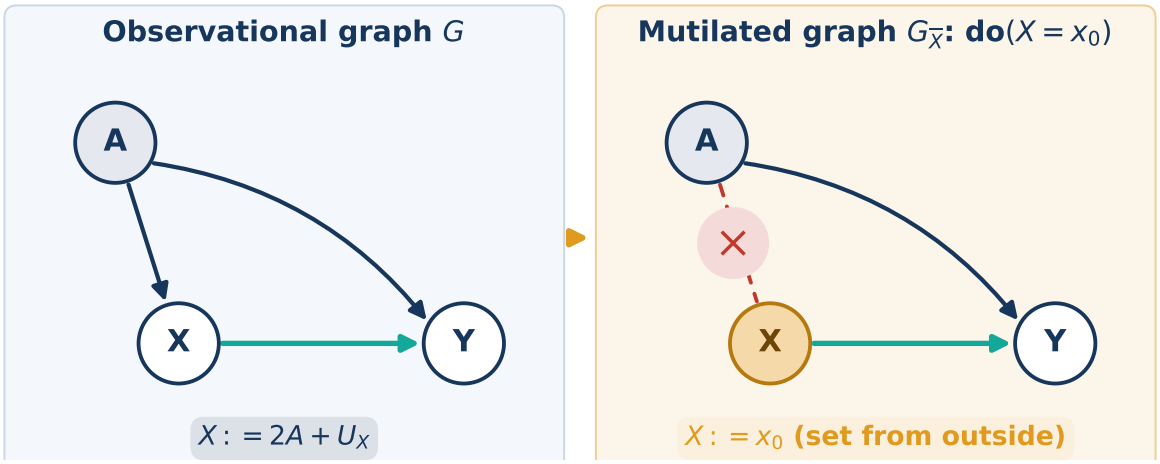


Figure 13.5. Graph surgery: $\mathrm{do}(X = x_0)$ deletes X's incoming edges (here, from confounder A) while keeping its outgoing edges to Y.

Theorem 13.3 (Truncated factorization)**.** *Let M be a Markovian SCM over $\mathbf{V} = \{V_1, \ldots, V_n\}$ and let $X = V_k$ be one of its variables. For every configuration $(v_1, \ldots, v_n)$,*

$$P(v_1, \ldots, v_n \mid \mathrm{do}(X = x)) = \begin{cases} \prod_{i \neq k} P\big(v_i \mid \mathrm{pa}(v_i)\big) & \text{if } v_k = x, \\ 0 & \text{if } v_k \neq x, \end{cases}$$

i.e. the configuration must agree with the intervention, and any parent value $\mathrm{pa}(v_i)$ that involves X is evaluated at $X = x$. In words: drop the factor $P(x \mid \mathrm{pa}(x))$ from the Bayesian-network factorization of Theorem 13.1 (Session 6) and fix $X = x$ everywhere else.

Proof. M_x is again a Markovian SCM (its noises are a subset of the original independent noises), with graph $G_{\overline{X}}$. By Theorem 13.1, $P_{M_x}(\mathbf{v}) = \prod_i P_{M_x}(v_i \mid \mathrm{pa}_i)$. For $V_i \neq X$ the mechanism and noise are unchanged, so $P_{M_x}(v_i \mid \mathrm{pa}_i) = P(f_i(\mathrm{pa}_i, U_i) = v_i) = P(v_i \mid \mathrm{pa}_i)$; the factor of X is the point mass $\mathbb{1}(v_X = x)$. (The kernels $P(v_i \mid \mathrm{pa}_i)$ are defined by the mechanisms for every parent value, so no positivity assumption is needed in the model; *estimating* them from data does require the relevant parent configurations to occur.) □

This is the single most important formula of this session, and it is the bridge connecting the entire directed-graphical-model machinery of Sessions 6–7 directly to causal queries: computing an interventional distribution is *exactly* an ordinary Bayesian-network computation (Session 7's variable elimination applies immediately) on the *mutilated* graph, using the very same CPTs as the original network for every equation except X's. No new algorithm is needed — only a different graph.

INTUITION

Contrast conditioning with intervening using our confounding example. Conditioning on $X = x$ (Session 12's $P(y \mid x)$) restricts attention to a subpopulation — everyone in that subpopulation still got their X value through the ordinary mechanism $X := 2A + U_X$, so if X is high partly because A is high, that information about A leaks into what we know about Y. Intervening, $\mathrm{do}(X = x)$, severs this: every unit is forced to $X = x$ regardless of its A, so the leak is cut at the source, exactly as Theorem 13.3 formalizes by simply deleting X's factor.

Summing the truncated factorization over all variables other than Y and $\mathbf{Z} = \mathrm{pa}(X)$ gives the **adjustment formula**

$$P(y \mid \mathrm{do}(x)) = \sum_{\mathbf{z}} P(y \mid x, \mathbf{z})\, P(\mathbf{z}) \qquad (\mathbf{Z} = \mathrm{pa}(X),\ Y \notin \mathrm{pa}(X)).$$

Derivation (assuming $P(x \mid \mathbf{z}) > 0$): for every configuration $\mathbf{v}$ with $v_k = x$, Theorem 13.3 says $P(\mathbf{v} \mid \mathrm{do}(x)) = P(\mathbf{v})/P(x \mid \mathbf{z})$, because the full product $P(\mathbf{v})$ differs from the truncated one only by the factor $P(x \mid \mathbf{z})$. Summing over all variables except Y and $\mathbf{Z}$ (the denominator depends only on $\mathbf{z}$) gives $P(y, \mathbf{z} \mid \mathrm{do}(x)) = P(y, x, \mathbf{z})/P(x \mid \mathbf{z}) = P(y \mid x, \mathbf{z})P(\mathbf{z})$; finally sum over $\mathbf{z}$. For the wage SCM this reads $P(y \mid \mathrm{do}(x)) = \int p(y \mid x, a)\, p(a)\, da$, and for the discrete examples below it is a two-term sum over the confounder. A *soft* intervention replaces f_X by a new mechanism such as a policy $\pi(x \mid \mathbf{z})$, which simply replaces the factor $P(x \mid \mathrm{pa}_x)$ — the way every RL policy acts on its environment.

PART III · INTERVENTIONS

13.7 Counterfactuals

Interventions ask "what would happen to a population if we set $X = x$?" Counterfactuals ask a sharper, unit-specific question: "for this specific unit, who we observed with $X = x$ and $Y = y$, what would Y have been had X been x' instead?" SCMs answer this via a three-step procedure:

1. **Abduction**: given the observed evidence e, replace the prior $P(\mathbf{U})$ by the posterior $P(\mathbf{U} \mid e)$. If every endogenous variable of the unit is observed *and* each mechanism is invertible in its own noise (e.g. additive noise, $V_i := g_i(\mathrm{pa}_i) + U_i$, so $U_i = V_i - g_i(\mathrm{pa}_i)$), the posterior is a point mass and $\mathbf{U}$ is pinned down exactly; otherwise it is a genuine distribution (Worked Examples 13.9 and 13.14).

2. **Action**: perform $\mathrm{do}(X = x')$ (graph surgery, Definition 13.4) on the model.
3. **Prediction**: compute Y in this modified model, using the *same* (abduced) value of $\mathbf{U}$ as this specific unit.

Definition 13.5 (Counterfactual)**.** For a unit $\mathbf{u}$, the *potential response* $Y_x(\mathbf{u})$ is the value of Y in the submodel M_x evaluated at $\mathbf{U} = \mathbf{u}$. For evidence e (an observed event about the endogenous variables), the counterfactual distribution is $P(Y_{x'} = y' \mid e) = \sum_{\mathbf{u}} \mathbb{1}\{Y_{x'}(\mathbf{u}) = y'\}\, P(\mathbf{u} \mid e)$ (an integral against $P(\mathbf{u} \mid e)$ for continuous noise). Two immediate consequences (Pearl 2009, ch. 7) are the *consistency* rule, $X(\mathbf{u}) = x \Rightarrow Y_x(\mathbf{u}) = Y(\mathbf{u})$ (setting X to the value it would have taken anyway changes nothing), and the *composition* rule, $W_x(\mathbf{u}) = w \Rightarrow Y_{x,w}(\mathbf{u}) = Y_x(\mathbf{u})$ (additionally setting W to the value it takes under $\mathrm{do}(x)$ changes nothing).

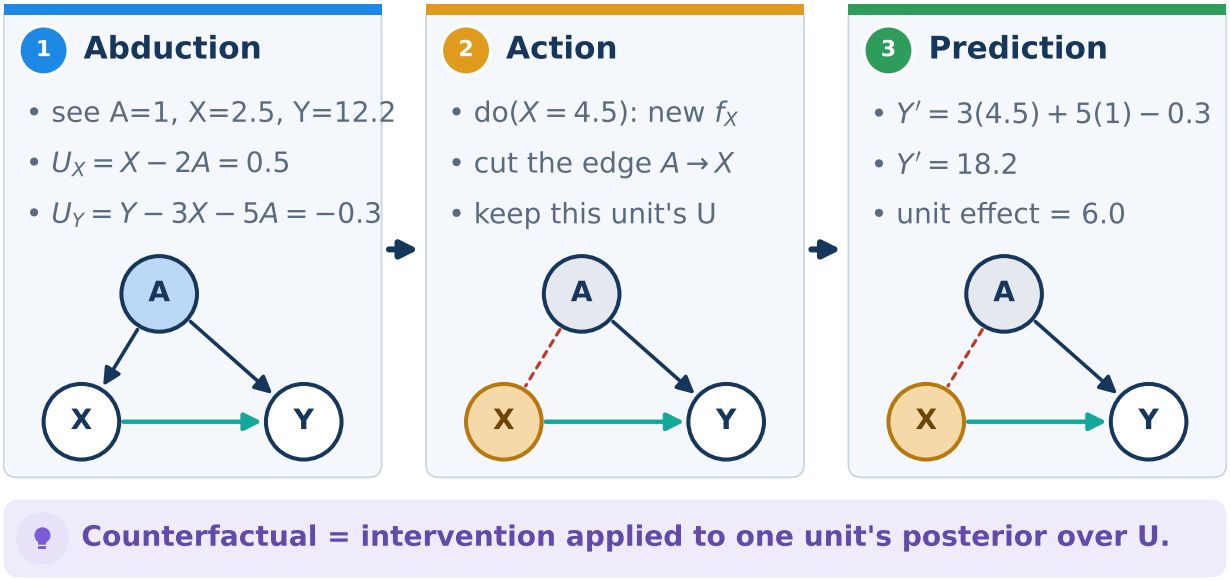


Figure 13.6. The three-step counterfactual computation on the wage SCM, for the individual of Worked Example 13.11.

Remark 13.1 (Bridge to potential outcomes). Students arriving from the potential-outcomes (Neyman–Rubin) tradition will recognize $Y_x(u)$ — "the value Y would take for unit u if X were set to x" — as exactly the SCM counterfactual just defined, usually written $Y(x)$ or Y_x. The two frameworks are provably equivalent in what they can express; SCMs simply make the generative mechanism explicit, which is convenient for reading identifiability directly off a graph (Session 14), while potential outcomes are often more convenient for stating assumptions directly in terms of estimands (Session 15). We will use both notations as convenient from here on.

13.7.1 Probabilities of causation

For binary X, Y, write x for the presence of the cause ($X = 1$), x' for its absence ($X = 0$), and y for the event $Y = 1$. Three counterfactual quantities capture "did X cause Y?" (Pearl 2009, ch. 9):

$$\mathrm{PN} = P(Y_{x'} = 0 \mid X = x, Y = 1), \quad \mathrm{PS} = P(Y_x = 1 \mid X = x', Y = 0), \quad \mathrm{PNS} = P(Y_x = 1, Y_{x'} = 0).$$

PN (probability of necessity) is the legal "but-for" question: among cases with the cause and the effect, how often would the effect have been absent without the cause? PS (probability of

sufficiency) asks the converse: among cases with neither cause nor effect, how often would the cause have produced the effect? PNS (probability of necessity and sufficiency) is the population share of units for which the effect occurs with the cause and not without it.

Proposition 13.2 (Bounds and monotone identification)**.** *Let* $p_1 = P(y \mid \mathrm{do}(x))$ *and* $p_0 = P(y \mid \mathrm{do}(x'))$*. Then* $\max(0, p_1 - p_0) \le \mathrm{PNS} \le \min(p_1, 1 - p_0)$*. If* Y *is* monotone *in* X *($Y_{x'} = 1 \Rightarrow Y_x = 1$ for every unit), then* $\mathrm{PNS} = p_1 - p_0$*; if moreover* X *is exogenous* $(Y_x, Y_{x'} \perp\!\!\!\perp X)$*, then* $\mathrm{PN} = (p_1 - p_0)/p_1$ *and* $\mathrm{PS} = (p_1 - p_0)/(1 - p_0)$*.*

Proof. Each unit has one of four *response types* $(Y_x, Y_{x'})$: always $(1,1)$, helped $(1,0)$, hurt $(0,1)$, never $(0,0)$, with probabilities a, h, k, n. Then $p_1 = a + h$, $p_0 = a + k$ and $\mathrm{PNS} = h$. From $h = p_1 - p_0 + k \ge p_1 - p_0$ and $h \ge 0$ we get the lower bound; from $h = p_1 - a \le p_1$ and $h \le 1 - a - k = 1 - p_0$ the upper bound. Monotonicity means $k = 0$, so $h = p_1 - p_0$. Under exogeneity the type distribution is the same in the $X = x$ and $X = x'$ groups; units with $X = x, Y = 1$ are of type always or helped (mass p_1), so $\mathrm{PN} = h/p_1$; units with $X = x', Y = 0$ are helped or never (mass $1 - p_0$), so $\mathrm{PS} = h/(1 - p_0)$. □

PART III · INTERVENTIONS

13.8 Which Way Does the Arrow Point? Additive-Noise Models

Worked Example 13.3 showed that for linear-Gaussian SCMs the direction of an edge is not identifiable from P. Surprisingly, Gaussianity is the *only* obstacle in the linear case. In an **additive-noise model** $Y := f(X) + U_Y$ with $U_Y \perp\!\!\!\perp X$, one fits regressions in both directions and checks which residual is independent of its regressor; for generic nonlinear f or non-Gaussian noise, only the causal direction admits an independent residual (Hoyer et al. 2009; Peters, Janzing and Schölkopf 2017). The linear non-Gaussian case is Theorem 13.6 in Part IV.

Worked Example 13.7 (Detecting direction with HSIC)**.** *Simulation setup* (`direction_data()`, `default_rng(3)`): $n = 300$ draws of $X := U_X$, $Y := X + U_Y$ with independent noises, either uniform on $[-\sqrt{3}, \sqrt{3}]$ (mean 0, variance $(2\sqrt{3})^2/12 = 1$) or $N(0,1)$. For each direction we fit the least-squares line ("$X \to Y$": regress Y on X; "$Y \to X$": regress X on Y) and test whether the residual is independent of the regressor with the Hilbert–Schmidt independence criterion (HSIC: Gaussian kernels with median-heuristic bandwidth; p-value from 200 random permutations of the residual, `default_rng(0)`). Results at level 5%: with uniform noise the causal fit gives $p = 0.249$ (independence not rejected) and the anti-causal fit $p = 0.005$ (rejected) — only the true direction survives. With Gaussian noise, $p = 0.144$ and $p = 0.731$ — both directions survive, as Worked Example 13.3 predicts.

PYTHON LAB · `s13_examples.py` · `ex_direction_lingam()`

```
uniform  noise:  X->Y  HSIC = 0.00147 (p = 0.249);  Y->X  HSIC = 0.00879 (p
= 0.005)
gaussian noise:  X->Y  HSIC = 0.00152 (p = 0.144);  Y->X  HSIC = 0.00079 (p
= 0.731)
```

PART IV

Mathematical Statistics of SCMs

Push-forwards, linear SEMs, path tracing and identifiability

PART IV · MATH STATISTICS

13.9 The Distribution of an SCM

MATHEMATICAL STATISTICS · The induced law is a push-forward

Let $(\Omega, \mathcal{A}, \mathbb{P})$ carry $\mathbf{U}$ with law $P_{\mathbf{U}}$. A recursive SCM defines a *solution map* $\Gamma : \mathbf{u} \mapsto \mathbf{v}$ (computed in topological order). The observational law is the push-forward $P_{\mathbf{V}} = P_{\mathbf{U}} \circ \Gamma^{-1}$, i.e. $P_{\mathbf{V}}(B) = P_{\mathbf{U}}(\{\mathbf{u} : \Gamma(\mathbf{u}) \in B\})$; the interventional law is $P_{\mathbf{U}} \circ \Gamma_x^{-1}$ for the solution map Γ_x of M_x; and a counterfactual law is the push-forward of the *posterior* $P_{\mathbf{U}|e}$ through Γ_x. All three rungs are push-forwards of a law on $\mathbf{U}$ — only the map or the law changes.

Proposition 13.3 (Existence and uniqueness of solutions)**.** *If the graph of an SCM is acyclic, then for every* $\mathbf{u}$ *the system* $v_i = f_i(\mathrm{pa}_i, u_i)$*,* $i = 1, \dots, n$*, has exactly one solution* $\mathbf{v} = \Gamma(\mathbf{u})$*, and* Γ *is measurable whenever the* f_i *are.*

Proof. Take a topological order. V_1 has no parents, so $v_1 = f_1(u_1)$ is forced. If $v_1, \dots, v_{i-1}$ are uniquely determined (measurable in $\mathbf{u}$), then $\mathrm{pa}_i \subseteq \{v_1, \dots, v_{i-1}\}$ and $v_i = f_i(\mathrm{pa}_i, u_i)$ is uniquely determined and measurable as a composition. Induction completes the proof. (Cycles such as $X := Y$, $Y := X$ can destroy uniqueness.) □

PART IV · MATH STATISTICS

13.10 Linear SEMs, Path Tracing and Regression

A *linear structural equation model* (SEM) is an SCM $V_i := \sum_j b_{ij} V_j + U_i$ with $b_{ij} \neq 0$ only if $V_j \to V_i$, independent noises with $\mathrm{Var}(U_i) = \omega_i$. In matrix form $\mathbf{V} = B\mathbf{V} + \mathbf{U}$ with $\Omega = \mathrm{diag}(\omega_1, \dots, \omega_n)$.

Theorem 13.4 (Covariance of a linear SEM)**.** *If* G *is acyclic,* $I - B$ *is invertible,* $\mathbf{V} = (I - B)^{-1}\mathbf{U}$*, and*

$$\Sigma = \mathrm{Cov}(\mathbf{V}) = (I - B)^{-1}\, \Omega\, (I - B)^{-\top}.$$

Proof. In a topological order B is strictly lower triangular, so $I - B$ is unit lower triangular with $\det(I - B) = 1$; hence it is invertible and $\mathbf{V} = B\mathbf{V} + \mathbf{U}$ has the unique solution $\mathbf{V} = (I - B)^{-1}\mathbf{U}$ (the linear case of Proposition 13.3). Then $\mathrm{Cov}(\mathbf{V}) = (I-B)^{-1}\mathrm{Cov}(\mathbf{U})(I-B)^{-\top} = (I-B)^{-1}\Omega(I-B)^{-\top}$. □

Theorem 13.5 (Trek rule; Wright's path tracing)**.** *A* trek *between* V_i *and* V_j *(*$i \neq j$*) is a pair of directed paths from a common* top *node* V_k*, one to* V_i *and one to* V_j *(a path is empty if*

$k \in \{i, j\}$); its weight is the product $\prod_{e\in\tau} b_e$ of the coefficients on both paths. A trek is simple *if its two paths share no node other than V_k — equivalently, it is a path between V_i and V_j without a collider. Then*

$$(a)\quad \Sigma_{ij} = \sum_{\text{all treks } \tau} \omega_{\mathrm{top}(\tau)} \prod_{e\in\tau} b_e, \qquad (b)\quad \Sigma_{ij} = \sum_{\text{simple treks } \tau} \mathrm{Var}\big(V_{\mathrm{top}(\tau)}\big) \prod_{e\in\tau} b_e.$$

Proof. (a) B is nilpotent ($B^n = 0$), so $(I-B)^{-1} = \sum_{m=0}^{n-1} B^m$. By induction on m, $(B^m)_{ik}$ is the sum, over directed paths of length m from V_k to V_i, of the product of their edge coefficients. Hence $\Lambda := (I-B)^{-1}$ has $\Lambda_{ik} = \sum_{\pi: V_k \rightsquigarrow V_i} \prod_{e\in\pi} b_e$ (with $\Lambda_{ii} = 1$). By Theorem 13.4, $\Sigma_{ij} = \sum_k \Lambda_{ik}\,\omega_k\,\Lambda_{jk}$: a sum over top nodes k of ω_k times the weight of a directed path $k \rightsquigarrow i$ and a directed path $k \rightsquigarrow j$, i.e. (a).

(b) Given a trek ($\pi_1 : k \rightsquigarrow i,\ \pi_2 : k \rightsquigarrow j$), let t be the *last* node of π_1 that also lies on π_2. Cut both paths at t: the two initial pieces are a pair of directed paths $k \rightsquigarrow t$, $k \rightsquigarrow t$, and the two remainders $t \rightsquigarrow i$, $t \rightsquigarrow j$ form a *simple* trek with top t (the remainder of π_1 meets π_2 only at t by the choice of t, and the remainder of π_2 cannot meet the part of π_1 before t, since that would create a directed cycle). Conversely, gluing any such pair of paths into t onto any simple trek with top t gives a trek between V_i and V_j, so the decomposition is a bijection. Grouping the terms of (a) by t therefore gives $\sum_t \sum_{\text{simple treks with top } t} \big(\sum_k \Lambda_{tk}^2 \omega_k\big) \prod b_e$, and $\sum_k \Lambda_{tk}\,\omega_k\,\Lambda_{tk} = \Sigma_{tt} = \mathrm{Var}(V_t)$ by Theorem 13.4. □

For standardized variables ($\mathrm{Var}(V_i) = 1$), form (b) is Sewall Wright's (1921) original rule: *a correlation equals the sum over all collider-free paths of the products of the path coefficients.*

Worked Example 13.8 (Tracing treks in a four-variable SEM)**.** Take $A := U_A$, $X := aA + U_X$, $M := mX + U_M$, $Y := cA + bX + dM + U_Y$ with $a = 0.8$, $m = 0.7$, $c = 0.5$, $b = 0.3$, $d = 0.6$ and all noise variances $\omega = 1$ (Fig. 13.7, left). We compute $\mathrm{Cov}(X, Y)$ with form (b) of Theorem 13.5. The variances of the two possible tops are $\mathrm{Var}(A) = 1$ and $\mathrm{Var}(X) = a^2\mathrm{Var}(A) + 1 = 0.64 + 1 = 1.64$. The simple treks between X and Y are:

1. top A, paths $A \to X$ and $A \to Y$: $\mathrm{Var}(A)\,a\,c = 1 \cdot 0.8 \cdot 0.5 = 0.4000$;
2. top X, paths (empty) and $X \to Y$: $\mathrm{Var}(X)\,b = 1.64 \cdot 0.3 = 0.4920$;
3. top X, paths (empty) and $X \to M \to Y$: $\mathrm{Var}(X)\,m\,d = 1.64 \cdot 0.7 \cdot 0.6 = 0.6888$.

So $\mathrm{Cov}(X, Y) = 0.4000 + 0.4920 + 0.6888 = 1.5808$, which matches entry (X, Y) of $(I-B)^{-1}(I-B)^{-\top}$ (Theorem 13.4). The *total causal effect* of X on Y is the sum over directed paths only, $b + md = 0.3 + 0.42 = 0.72$; the OLS slope of Y on X is $\mathrm{Cov}(X, Y)/\mathrm{Var}(X) = 1.5808/1.64 = 0.9639$, biased by the confounding trek divided by $\mathrm{Var}(X)$: $ac/\mathrm{Var}(X) = 0.4/1.64 = 0.2439 = 0.9639 - 0.72$. The population regressions in the pylab box show how the coefficient of X changes with the regressor set: 0.9639 on X alone, 0.72 (total effect) once A is added, and 0.3 (direct effect b) once A and M are added.

PYTHON LAB · `s13_examples.py` · `ex_path_tracing()`

```
per trek:  X<-A->Y 0.4000;  X->Y 0.4920;  X->M->Y 0.6888
trek rule:  Cov(X,Y) = 0.40 + 1.64*0.72 = 1.5808;  matrix:  Cov(X,Y) =
1.5808
regress Y on X: [0.9639];  on X, A: [0.72 0.5];  on X, A, M: [0.3 0.5 0.6]
OLS slope = 1.5808/1.64 = 0.9639;  OVB: a c / Var(X) = 0.4/1.64 = 0.2439
```

(Population regressions are computed from Σ; a simulation with $n = 50\,000$, `default_rng(21)`, reproduces every entry of Σ to within 0.0121.)

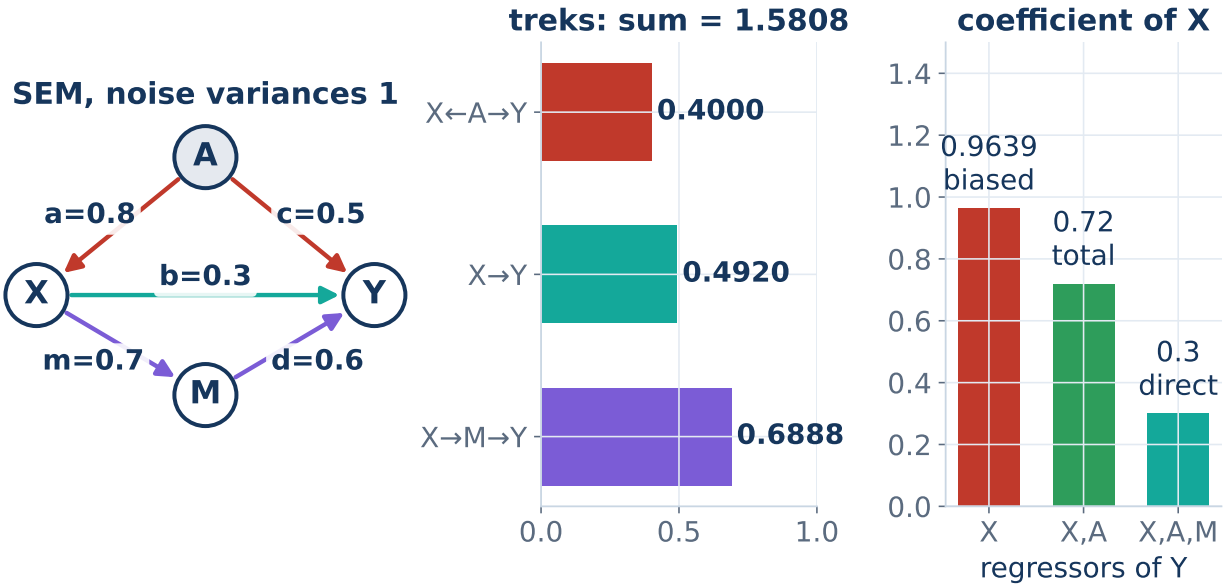


Figure 13.7. Path tracing in the linear SEM of Worked Example 13.8. Left: the graph with its structural coefficients. Middle: $\mathrm{Cov}(X, Y) = 1.5808$ as the sum of the three simple-trek contributions $0.4000 + 0.4920 + 0.6888$ (form (b) of Theorem 13.5). Right: the population coefficient of X depends on what else we regress on — 0.9639 (no adjustment, biased), 0.72 (adjust for the confounder A: total effect), 0.3 (adjust for A and the mediator M: direct effect).

Proposition 13.4 (Omitted-variable bias)**.** *Let $Y := \mathbf{b}^\top \mathbf{X} + \mathbf{c}^\top \mathbf{A} + U_Y$ with $U_Y \perp\!\!\!\perp (\mathbf{X}, \mathbf{A})$ and Σ_{XX} invertible. The population least-squares coefficient of Y on $\mathbf{X}$ alone is*

$$\boldsymbol{\beta} = \Sigma_{XX}^{-1}\Sigma_{XY} = \mathbf{b} + \Sigma_{XX}^{-1}\Sigma_{XA}\,\mathbf{c}.$$

In the scalar case with $X := aA + U_X$, $\mathrm{Var}(A) = \sigma_A^2$, $\mathrm{Var}(U_X) = \sigma_X^2$: $\beta = b + ac\sigma_A^2/(a^2\sigma_A^2 + \sigma_X^2)$.

Proof. $\Sigma_{XY} = \mathrm{Cov}(\mathbf{X}, \mathbf{b}^\top\mathbf{X} + \mathbf{c}^\top\mathbf{A} + U_Y) = \Sigma_{XX}\mathbf{b} + \Sigma_{XA}\mathbf{c} + \mathbf{0}$; multiply by Σ_{XX}^{-1}. In the scalar case $\Sigma_{XA} = a\sigma_A^2$ and $\Sigma_{XX} = a^2\sigma_A^2 + \sigma_X^2$. □

The bias term $\Sigma_{XX}^{-1}\Sigma_{XA}\mathbf{c}$ is the coefficient of the auxiliary regression of $\mathbf{A}$ on $\mathbf{X}$ times the effect of $\mathbf{A}$ on Y: it vanishes iff the omitted variables are uncorrelated with $\mathbf{X}$ or irrelevant for Y — the algebraic face of "confounding needs both arrows of the fork".

Proposition 13.5 (Regression on the parents recovers the mechanism)**.** *In a Markovian linear SEM with $\Omega \succ 0$, the population regression of V_i on $\mathrm{pa}(V_i)$ has coefficients $(b_{ij})_{j \in \mathrm{pa}(i)}$ and residual variance ω_i.*

Proof. $V_i = \mathbf{b}_i^\top \mathrm{pa}_i + U_i$ with $U_i \perp\!\!\!\perp \mathrm{pa}_i$ by Theorem 13.1 (parents are non-descendants). Hence $\mathrm{Cov}(\mathrm{pa}_i, V_i - \mathbf{b}_i^\top \mathrm{pa}_i) = \mathrm{Cov}(\mathrm{pa}_i, U_i) = \mathbf{0}$, which are the normal equations; they have a unique solution because $\Sigma_{\mathrm{pa}_i \mathrm{pa}_i}$ is a principal submatrix of $\Sigma \succ 0$ (Theorem 13.4 with $\Omega \succ 0$). The residual is U_i, with variance ω_i. □

MATHEMATICAL STATISTICS · Regression coefficients versus structural coefficients

A regression coefficient belongs to P *and the chosen regressor set*; a structural coefficient belongs to a mechanism. They coincide for the parent set (Proposition 13.5). The coefficient of X is the *total* effect when the other regressors block every back-door path and contain no descendant of X (here $\{A\}$: 0.72), and the *direct* effect when they also block mediated paths ($\{A, M\}$: 0.3); colliders or descendants of Y bias it again. Choosing regressors is a causal decision — Session 14's back-door criterion.

Theorem 13.6 (LiNGAM identifiability; Shimizu et al. 2006)**.** *If* $\mathbf{V} = B\mathbf{V} + \mathbf{U}$ *with acyclic* B *and independent noises of which at most one is Gaussian, then* B *(hence the causal DAG) is identifiable from* $P(\mathbf{V})$*.*

Proof for two variables. We use the Darmois–Skitovich theorem: if $L_1 = \sum \alpha_k U_k$ and $L_2 = \sum \beta_k U_k$ are independent linear forms in independent U_k, every U_k with $\alpha_k \beta_k \neq 0$ is Gaussian. Suppose $Y := bX + U_Y$, $X \perp\!\!\!\perp U_Y$, $b \neq 0$, and try the reverse model: the residual of regressing X on Y is $R = X - \gamma Y = (1 - \gamma b)X - \gamma U_Y$ with $\gamma = b\mathrm{Var}(X)/\mathrm{Var}(Y) \neq 0$ and $1 - \gamma b = \mathrm{Var}(U_Y)/\mathrm{Var}(Y) > 0$. Both R and $Y = bX + U_Y$ load on X and on U_Y with nonzero coefficients, so $R \perp\!\!\!\perp Y$ would force X and U_Y to be Gaussian. Hence if either is non-Gaussian, the reverse model fails and the direction is identified. □

Theorem 13.7 (Counterfactuals without effect heterogeneity)**.** *Suppose* $Y_x(\mathbf{u}) = \tau x + g(\mathbf{u})$ *for all* $x, \mathbf{u}$ *(constant effect* τ*, e.g. any linear SEM with additive noise). Then for any evidence* e *that includes* $X = x$*, the counterfactual law of* $Y_{x'}$ *is the factual law of* Y *given* e *shifted by* $\tau(x' - x)$*: in particular* $\mathbb{E}[Y_{x'} \mid e] = \mathbb{E}[Y \mid e] + \tau(x' - x)$*.*

Proof. For every $\mathbf{u}$ consistent with e, consistency gives $Y(\mathbf{u}) = Y_x(\mathbf{u})$, so $Y_{x'}(\mathbf{u}) = \tau x' + g(\mathbf{u}) = Y(\mathbf{u}) + \tau(x' - x)$. Push the posterior $P(\mathbf{u} \mid e)$ forward through both sides. □

Theorem 13.7 explains why abduction looked trivial in the linear wage example: the unit-level effect does not depend on $\mathbf{u}$. With heterogeneity it does, and abduction matters.

MATHEMATICAL STATISTICS · Gaussian abduction with effect heterogeneity

Let $A \sim N(0,1)$ be *unobserved*, $X := 2A + U_X$ and $Y := (3 + A)X + U_Y$, with $U_X, U_Y \sim N(0,1)$ independent of each other and of A. Given $X = x, Y = y$, both equations are linear in A: since $U_X = x - 2A$ and $U_Y = y - 3x - Ax$, the posterior density is proportional to $\exp\{-a^2/2\} \exp\{-(x - 2a)^2/2\} \exp\{-(y - 3x - ax)^2/2\}$. The exponent is $-\frac{1}{2}\big[(1 + 4 + x^2)a^2 - 2a\,(2x + x(y - 3x))\big] + \text{const}$, so completing the square, $A \mid (x, y) \sim N(m, v)$ with

$$v = \frac{1}{1 + 4 + x^2}, \qquad m = v\,\big(2x + x(y - 3x)\big).$$

For the same unit U_Y is unchanged, so $Y_{x'} = (3 + A)x' + U_Y$ and $Y_{x'} - Y = (3 + A)(x' - x)$; hence $Y_{x'} \mid (x, y) \sim N\big(y + (x' - x)(3 + m),\ (x' - x)^2 v\big)$, whereas the population effect is $\mathbb{E}[Y \mid \mathrm{do}(x')] - \mathbb{E}[Y \mid \mathrm{do}(x)] = 3(x' - x)$.

Worked Example 13.9 (A unit that gains more than average)**.** For $x = 2$, $y = 9$, $x' = 3$:

$$v = \frac{1}{1+4+2^2} = \frac{1}{9} = 0.1111,$$
$$m = v\left(2\cdot 2 + 2\,(9 - 3\cdot 2)\right) = \frac{4+6}{9} = 1.1111,$$
$$\mathbb{E}[Y_3 \mid X{=}2, Y{=}9] = 9 + (3-2)(3+1.1111) = 13.1111,$$
$$\mathrm{sd}(Y_3 \mid X{=}2, Y{=}9) = |3-2|\sqrt{1/9} = 0.3333.$$

High income for the education reveals high ability, so this unit's gain ($13.1111 - 9 = 4.1111$) exceeds the average effect ($3\cdot(3-2) = 3$). *Monte Carlo check*: simulate $n = 4\times 10^6$ units (`default_rng(99)`), keep the 2475 with $|X-2| < 0.05$ and $|Y-9| < 0.15$, and evaluate $Y_3 = (3+A)\cdot 3 + U_Y$ for each: the mean is 13.101.

PYTHON LAB · `s13_examples.py` · `ex_hetero_counterfactual()`

```
posterior A | x, y :   mean = 1.1111, var = 0.1111, sd = 0.3333
E[Y_(x'=3) | X=2, Y=9] = 13.1111,  sd = 0.3333
Monte Carlo (n=4e6, seed=99; 2475 units kept):  mean Y' = 13.101
```

PART IV · MATH STATISTICS

13.11 Worked Examples

Worked Example 13.10 (The wage SCM: observational vs. interventional slope)**.** Using Example 13.1, compute (i) the slope of the observational regression of Y on X, and (ii) the true causal effect $\frac{d}{dx}\mathbb{E}[Y \mid \mathrm{do}(X = x)]$.

Solution. Substituting $X = 2A + U_X$ into $Y = 3X + 5A + U_Y$ gives $Y = 6A + 3U_X + 5A + U_Y = 11A + 3U_X + U_Y$. With unit-variance independent noises, $\mathrm{Cov}(X, Y) = \mathrm{Cov}(2A + U_X,\ 11A + 3U_X + U_Y) = 22\,\mathrm{Var}(A) + 3\,\mathrm{Var}(U_X) = 22 + 3 = 25$, and $\mathrm{Var}(X) = 4\,\mathrm{Var}(A) + \mathrm{Var}(U_X) = 4 + 1 = 5$, giving observational slope $25/5 = 5$. For the interventional quantity, Theorem 13.3 says: drop X's equation, fix $X = x$, keep $A := U_A$ and $Y := 3X + 5A + U_Y$ unchanged; then $\mathbb{E}[Y \mid \mathrm{do}(x)] = 3x + 5\mathbb{E}[A] + \mathbb{E}[U_Y] = 3x$ (since $\mathbb{E}[A] = \mathbb{E}[U_Y] = 0$), so $\frac{d}{dx}\mathbb{E}[Y \mid \mathrm{do}(x)] = 3$ exactly — matching the structural coefficient by construction, and matching Session 12's warning precisely: the naive association (5) overstates the true causal effect (3) by exactly the confounding contribution $ac/(a^2+1) = 2\cdot 5/(4+1) = 2$, with $a = 2$, $c = 5$ from Example 13.1 (Proposition 13.4; see also Section 13.14). Figure 13.8 confirms both values on simulated data (`simulate_wage()`, `interventional_curve()`): an ordinary least-squares regression of Y on X for $n = 2000$ draws of the SCM (`default_rng(123)`) has slope 5.03, while directly simulating the intervention — at each of 30 equally spaced values x spanning the observed range of X, set $X := x$ and average $Y = 3x + 5A + U_Y$ over 5000 fresh draws of (A, U_Y) (`default_rng(7)`) — and fitting a line through the 30 averages gives slope 3.00.

PYTHON LAB · `s13_examples.py` · `ex_wage_slopes()`

```
exact:  Cov(X,Y) = 25, Var(X) = 5, slope = 5.00;  d/dx E[Y | do(X=x)] = 3
simulation (n=2000, seed=123):  observational slope = 5.03
```

```
simulation of do(X=x) on 30 grid points, 5000 draws each (seed=7):
interventional slope = 3.00
bias formula a*c/(a^2+1) = 2*5/(4+1) = 2.00;  b + bias = 5.00
```

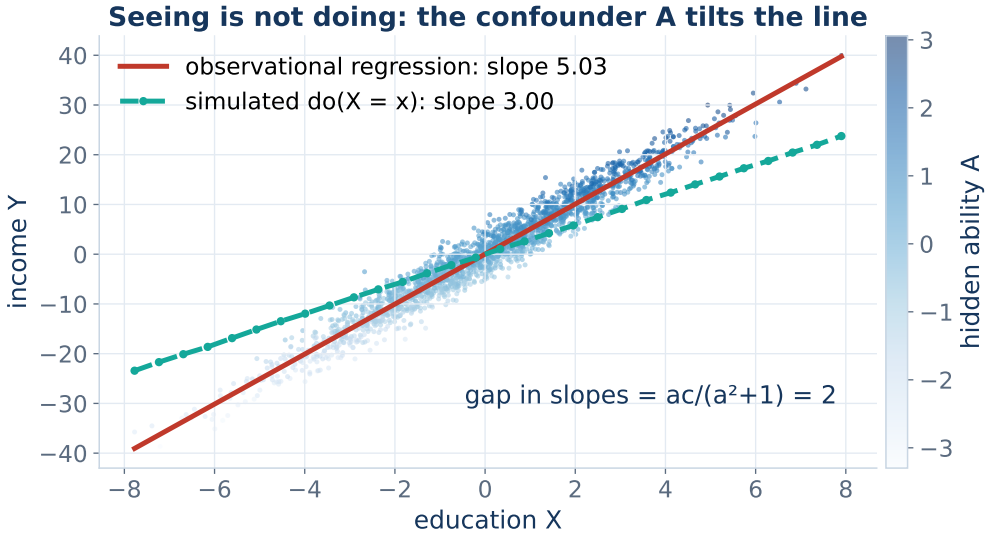


Figure 13.8. Simulated data from the wage SCM ($n = 2000$ points, coloured by the hidden ability A): the observational regression line (slope 5.03) is biased upward relative to the simulated interventional curve $\mathbb{E}[Y \mid \mathrm{do}(X = x)]$ (slope 3.00); the gap is $ac/(a^2 + 1) = 2$ (Worked Example 13.10).

Worked Example 13.11 (A numeric counterfactual)**.** In the wage SCM of Example 13.1, suppose that for one specific individual we observe all three endogenous variables: ability $A = 1$ (for this illustration we assume a test measured it), education $X = 2.5$ and income $Y = 12.2$. *Counterfactual question*: what would this same person's income have been with 2 additional years of education, i.e. $X' = 2.5 + 2 = 4.5$, *holding their personal ability and personal luck (U_Y) fixed*?

1. *Abduction.* Each equation is additive in its noise, so we invert it: $U_A = A = 1$, $U_X = X - 2A = 2.5 - 2 = 0.5$, $U_Y = Y - 3X - 5A = 12.2 - 7.5 - 5 = -0.3$.
2. *Action.* Replace $X := 2A + U_X$ by $X := 4.5$; keep $A := U_A$ and $Y := 3X + 5A + U_Y$.
3. *Prediction.* $Y' = 3(4.5) + 5(1) + (-0.3) = 13.5 + 5 - 0.3 = 18.2$.

The counterfactual effect for this individual is $18.2 - 12.2 = 6.0$, exactly 3×2 additional years — because the structural coefficient on X is homogeneous (3) across individuals in this simple linear model (Theorem 13.7); in a model with individual-level effect heterogeneity (coefficients depending on A or U), different individuals could have different counterfactual gains (Worked Example 13.9), a theme central to Session 15's treatment-effect heterogeneity.

PYTHON LAB · s13_examples.py · ex_counterfactual_individual()

```
abduction:  U_X = X - 2A = 0.5,  U_Y = Y - 3X - 5A = 12.2 - 7.5 - 5 = -0.3
factual:  X = 2.5, Y = 12.2;  counterfactual X' = 4.5:  Y' = 18.2,
individual effect = 6.0
```

Worked Example 13.12 (Revisiting kidney stones as an SCM)**.** Model Session 12's kidney-stone study (Charig et al. 1986; 700 patients) with Z (stone size: small or large), X (treatment:

A = open surgery, B = percutaneous nephrolithotomy), R (recovery), and structural equations $Z := U_Z$, $X := g(Z, U_X)$ (physicians' choice depends on stone size) and $R := h(X, Z, U_R)$ (recovery depends on both treatment and stone size). The naive comparison $P(R{=}1 \mid X)$ mixes Z's effect on R with X's effect via the same fork logic as the wage example. By Theorem 13.3, $\mathrm{do}(X = x)$ drops the factor $P(x \mid z)$ and keeps Z's marginal law, which gives the adjustment formula $P(R{=}1 \mid \mathrm{do}(x)) = \sum_z P(R{=}1 \mid x, z)P(z)$.
Data (real counts, recoveries/patients): A small $81/87 = 0.9310$, A large $192/263 = 0.7300$; B small $234/270 = 0.8667$, B large $55/80 = 0.6875$. Stone-size marginal: $P(\text{small}) = (87 + 270)/700 = 357/700 = 0.5100$, $P(\text{large}) = (263 + 80)/700 = 343/700 = 0.4900$.
Naive: $P(R{=}1 \mid A) = (81 + 192)/(87 + 263) = 273/350 = 0.7800$, $P(R{=}1 \mid B) = (234 + 55)/(270 + 80) = 289/350 = 0.8257$ — B looks better.
Interventional:

$$P(R{=}1 \mid \mathrm{do}(A)) = 0.9310 \cdot 0.5100 + 0.7300 \cdot 0.4900 = 0.8325,$$
$$P(R{=}1 \mid \mathrm{do}(B)) = 0.8667 \cdot 0.5100 + 0.6875 \cdot 0.4900 = 0.7789,$$

so A is better by 0.0537 (unrounded difference) once the case mix is held at the population's. This is the stratify-by-size tabulation of Julious and Mullee (1994) used in Session 12, now justified by a theorem; Session 14 shows that $\{Z\}$ is a valid adjustment set even when the mechanisms g, h are unknown (the back-door criterion).

PYTHON LAB · s13_examples.py · ex_kidney_preview()

```
treatment A: P(R|T,small) = 81/87 = 0.9310, P(R|T,large) = 192/263 = 0.7300
treatment B: P(R|T,small) = 234/270 = 0.8667, P(R|T,large) = 55/80 = 0.6875
treatment A: naive P(R|T) = 0.7800  adjusted P(R|do(T)) = 0.8325
treatment B: naive P(R|T) = 0.8257  adjusted P(R|do(T)) = 0.7789
P(small) = 0.5100, P(large) = 0.4900;  adjusted difference A - B = 0.0537
```

Additional Worked Examples (Instructor Supplement)

Worked Example 13.13 (Truncated factorization on a discrete (non-Gaussian) SCM)**.** Theorem 13.3 applies to any discrete structural equations implemented via CPTs, not only to linear-Gaussian SCMs. Let A (market boom) have $P(A{=}1) = 0.3$; let advertising campaign X have $P(X{=}1 \mid A{=}1) = 0.9$, $P(X{=}1 \mid A{=}0) = 0.2$; and let high sales Y have $P(Y{=}1 \mid X{=}1, A{=}1) = 0.9$, $P(Y{=}1 \mid X{=}1, A{=}0) = 0.6$, $P(Y{=}1 \mid X{=}0, A{=}1) = 0.7$, $P(Y{=}1 \mid X{=}0, A{=}0) = 0.2$. Compute the true causal effect of the campaign and compare it to the naive observational contrast.
Solution (interventional, via Theorem 13.3). Drop the factor $P(x \mid a)$ and fix X:

$$P(Y{=}1 \mid \mathrm{do}(X{=}1)) = \sum_a P(Y{=}1 \mid X{=}1, a)P(a) = 0.9(0.3) + 0.6(0.7) = 0.27 + 0.42 = \mathbf{0.69},$$

$$P(Y{=}1 \mid \mathrm{do}(X{=}0)) = 0.7(0.3) + 0.2(0.7) = 0.21 + 0.14 = \mathbf{0.35}.$$

True causal effect $= 0.69 - 0.35 = \mathbf{0.34}$.
Solution (observational). First $P(X{=}1) = 0.9(0.3) + 0.2(0.7) = 0.27 + 0.14 = 0.41$. By Bayes' rule, $P(A{=}1 \mid X{=}1) = 0.9(0.3)/0.41 \approx 0.6585$, and, using $P(X{=}0 \mid A{=}1) = 1 - 0.9 = 0.1$ and $P(X{=}0) = 1 - 0.41 = 0.59$, $P(A{=}1 \mid X{=}0) = 0.1(0.3)/0.59 \approx 0.0508$; hence $P(A{=}0 \mid X{=}1) =$

$1 - 0.6585 = 0.3415$ and $P(A{=}0 \mid X{=}0) = 1 - 0.0508 = 0.9492$. Conditioning on X keeps the factor $P(x \mid a)$, so $P(Y{=}1 \mid x) = \sum_a P(Y{=}1 \mid x, a)P(a \mid x)$:

$$P(Y{=}1 \mid X{=}1) \approx 0.9(0.6585) + 0.6(0.3415) \approx 0.5927 + 0.2049 = \mathbf{0.7976},$$

$$P(Y{=}1 \mid X{=}0) \approx 0.7(0.0508) + 0.2(0.9492) \approx 0.0356 + 0.1898 = \mathbf{0.2254}.$$

Naive difference $\approx 0.7976 - 0.2254 = \mathbf{0.5721}$ (computed from unrounded values), badly overstating the true effect (0.34) — the same confounding-bias phenomenon as Example 13.1, now for a fully categorical SCM where every quantity is an ordinary CPT rather than a Gaussian regression coefficient.

PYTHON LAB · s13_examples.py · ex_discrete_truncated()

```
P(Y=1|do(X=1)) = 0.69,  P(Y=1|do(X=0)) = 0.35,  causal effect = 0.34
P(X=1) = 0.41; P(A=1|X=1) = 0.6585; P(A=1|X=0) = 0.0508
P(Y=1|X=1) = 0.7976,  P(Y=1|X=0) = 0.2254,  naive difference = 0.5721
```

Worked Example 13.14 (Abduction is not always a single point: a non-invertible discrete mechanism)**.** In the wage counterfactual (Worked Example 13.11) abduction was exact and deterministic because every variable was observed and each structural equation was *invertible* in its noise given the parents. Worked Example 13.9 already lost the first condition (ability unobserved); here we lose the second. Let $X := \mathbb{1}[U_1{=}1 \text{ or } U_2{=}1]$, with U_1, U_2 i.i.d. Bernoulli(0.5), independent of everything else. Suppose we observe $X{=}1$ but never see U_1, U_2 individually. What does abduction give us?

Solution. The prior over (U_1, U_2) is uniform over 4 equally likely combinations, each probability 0.25. Three of the four combinations are consistent with $X{=}1$: $(1,0), (0,1), (1,1)$ (only $(0,0)$ gives $X{=}0$). Since all three had equal prior probability, each has posterior probability $0.25/0.75 = \mathbf{1/3}$ given $X{=}1$:

$$P(U_1{=}1, U_2{=}0 \mid X{=}1) = P(U_1{=}0, U_2{=}1 \mid X{=}1) = P(U_1{=}1, U_2{=}1 \mid X{=}1) = \tfrac{1}{3}.$$

Unlike Worked Example 13.11, abduction here does *not* pin down a single value of the exogenous noise — it yields a genuine three-point *distribution* over (U_1, U_2). A downstream counterfactual computation must then *average* over this posterior, rather than plug in one exact noise value as the prediction step did in Worked Example 13.11. For instance, "had the second cause been switched off, would X still have been 1?" is the counterfactual $X_{U_2=0} = \mathbb{1}[U_1{=}1]$ (set $U_2 := 0$, keep U_1), so its probability is $P(U_1{=}1 \mid X{=}1) = P(1,0 \mid X{=}1) + P(1,1 \mid X{=}1) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$. Non-invertible structural equations — common whenever a discrete variable is a many-to-one function of its causes — are the norm rather than the exception outside idealized linear models, and this is precisely why real implementations of counterfactual reasoning (e.g., in probabilistic programming systems) must track a full posterior over $\mathbf{U}$, not a point estimate.

PYTHON LAB · s13_examples.py · ex_noninvertible_abduction()

```
P(U1=0, U2=1 | X=1) = 0.3333;  P(U1=1, U2=0 | X=1) = 0.3333;  P(U1=1, U2=1 |
X=1) = 0.3333
P(X_(U2=0) = 1 | X=1) = 0.6667  (averaged over the abduced posterior)
```

PART V

SCMs Across Artificial Intelligence

World models, agents that ask “what if”, and credit among many agents

PART V · AI CONNECTIONS

13.12 SCMs in Agents, RL and Multi-Agent Systems

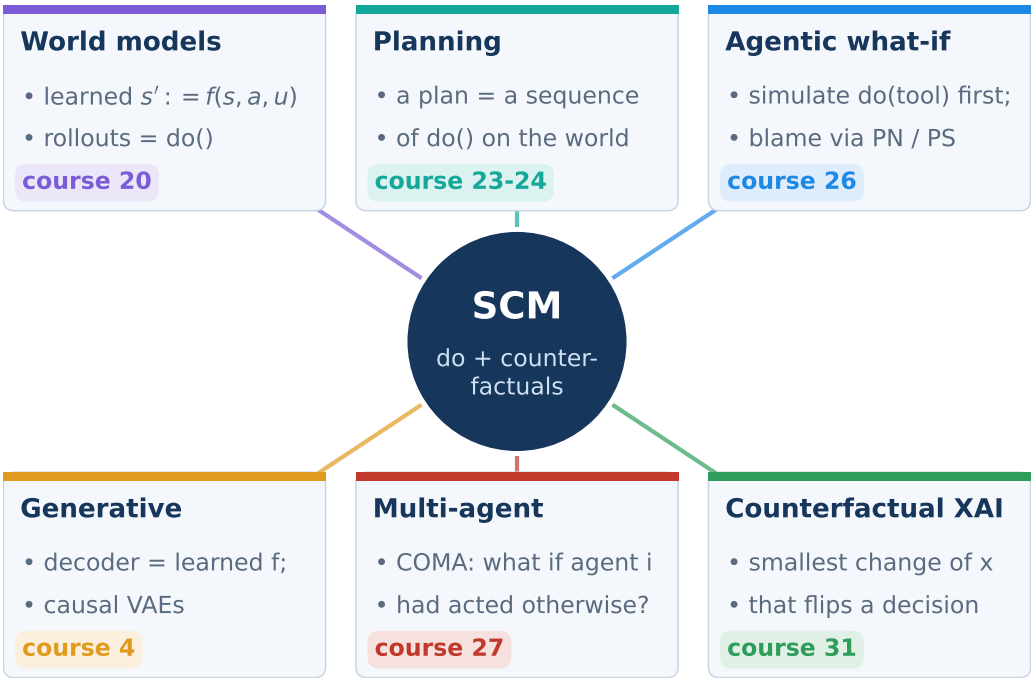


Figure 13.9. Where the structural-causal-model ideas of this session reappear in the Advanced AI course.

13.12.1 Agentic AI: what-if simulation and blame

An LLM agent deciding whether to call a costly tool is asking an interventional question about its environment, $P(\text{success} \mid \text{do}(\text{tool} = 1))$ versus $\text{do}(\text{tool} = 0)$; after the episode, a reflective agent (course session 26) asks a counterfactual one: *was the tool call necessary for this success?* That is exactly PN.

Worked Example 13.15 (Did calling the tool cause the success?)**.** From randomized A/B runs an agent platform knows $p_1 = P(S{=}1 \mid \text{do}(T{=}1)) = 0.80$ and $p_0 = P(S{=}1 \mid \text{do}(T{=}0)) = 0.30$ (illustrative numbers). By Proposition 13.2, $\max(0,\, 0.80 - 0.30) = 0.50 \le \text{PNS} \le \min(0.80,\, 1 - 0.30) = 0.70$. If the tool can never *hurt* (monotonicity) and tool use was randomized (exogeneity), $\text{PNS} = p_1 - p_0 = 0.50$, $\text{PN} = (p_1 - p_0)/p_1 = 0.5/0.8 = 0.6250$ and $\text{PS} = (p_1 - p_0)/(1 - p_0) = 0.5/0.7 = 0.7143$. Two SCMs with the same p_1, p_0 show the bounds are sharp (notation of the proof of Proposition 13.2: $p_1 = a + h$, $p_0 = a + k$, $\text{PNS} = h$): the monotone one has response types (always, helped, hurt, never) $= (0.30, 0.50, 0, 0.20)$, so

$p_1 = 0.30 + 0.50 = 0.80$, $p_0 = 0.30 + 0 = 0.30$ and PNS $= 0.50$; the non-monotone one has $(0.10, 0.70, 0.20, 0)$, so $p_1 = 0.10 + 0.70 = 0.80$, $p_0 = 0.10 + 0.20 = 0.30$ and PNS $= 0.70$. Experiments alone cannot tell them apart; the monotonicity assumption is a substantive claim about the tool.

PYTHON LAB · s13_examples.py · ex_probabilities_causation()

```
PNS bounds:  [max(0, 0.80-0.30), min(0.80, 1-0.30)] = [0.50, 0.70]
under monotonicity:  PNS = 0.50,  PN = (p1-p0)/p1 = 0.6250,  PS =
(p1-p0)/(1-p0) = 0.7143
monotone SCM     : always=0.30 helped=0.50 hurt=0.00 never=0.20 -> p1=0.80,
p0=0.30, PNS=0.50
non-monotone SCM: always=0.10 helped=0.70 hurt=0.20 never=0.00 -> p1=0.80,
p0=0.30, PNS=0.70
```

LINK TO COURSE SESSION 26 · Agentic AI

Planning calls do(·) on an explicit model of the environment before acting; reflection calls the counterfactual machinery after acting ("which step caused the failure?"). Logged traces are observational data generated by the agent's own policy — the confounding of Session 12 — so they answer rung-2 questions only after the identification analysis of Session 14.

13.12.2 Reinforcement learning: world models as SCMs

A model-based RL agent (course session 20) learns a transition model; written as $s_{t+1} := f(s_t, a_t, u_t)$, $r_t := g(s_t, a_t)$ with independent u_t, it is a Markovian SCM unrolled in time, and a policy is a soft intervention on the action nodes. It answers two different questions: the *interventional* expected cost of a new policy (fresh noise), and the *counterfactual* outcome of *this* logged episode under the new policy (same noise), the basis of counterfactual data augmentation. Planning (course sessions 23–24) is intervention on the same SCM: a STRIPS action schema is a structural assignment whose frame assumption ("everything else stays the same") is exactly the autonomy of the other mechanisms.

Worked Example 13.16 (Counterfactual replay in a linear world model)**.** Let $s_{t+1} := 0.9s_t + 0.5a_t + u_t$ for $t = 0, \dots, 5$, with $u_t \overset{\text{iid}}{\sim} N(0, 0.5^2)$, $s_0 = 2$ and episode cost $\sum_{t=1}^{6} s_t^2 + 0.1 \sum_{t=0}^{5} a_t^2$. A logged episode under the passive policy $a_t = 0$ (simulated with `default_rng(2024)`) visits

$$s = (2,\ 2.314,\ 2.904,\ 3.187,\ 2.382,\ 1.447,\ 1.336),$$

with cost $5.357 + 8.433 + 10.156 + 5.672 + 2.094 + 1.785 = 33.497$ (no action cost since $a_t = 0$).

1. *Abduction* is exact because the mechanism is invertible in u_t: $\hat{u}_t = s_{t+1} - 0.9s_t - 0.5a_t$, e.g. $\hat{u}_0 = 2.314 - 0.9 \cdot 2 = 0.514$; all six values are $\hat{u} = (0.514,\ 0.821,\ 0.573,\ -0.487,\ -0.696,\ 0.034)$.
2. *Action*: replace the policy by the feedback rule $a_t := -s_t$, so $s_{t+1} = 0.9s_t - 0.5s_t + \hat{u}_t = 0.4s_t + \hat{u}_t$.
3. *Prediction*: $s_1 = 0.4 \cdot 2 + 0.514 = 1.314$, $s_2 = 0.4 \cdot 1.314 + 0.821 = 1.347$, and so on: $s = (2,\ 1.314,\ 1.347,\ 1.112,\ -0.042,\ -0.713,\ -0.252)$. The state part of the cost is $1.728 + 1.814 + 1.237 + 0.002 + 0.509 + 0.063 = 5.352$ and the action part $0.1 \sum_{t=0}^{5} a_t^2 = 0.1 \sum_{t=0}^{5} s_t^2 = 0.929$, total 6.280 (sums computed from unrounded values).

In this episode the controller would have saved $33.497 - 6.280 = 27.2$ cost units. The interventional values average over fresh noise instead. Since $s_{t+1} = r\, s_t + u_t$ with $r = 0.9$ (passive) or $r = 0.4$ (feedback), $\mathbb{E}[s_t^2] = r^{2t} s_0^2 + 0.25\,(1 - r^{2t})/(1 - r^2)$, which gives exactly $\mathbb{E}[\text{cost} \mid \text{do}(\pi = \text{feedback})] = 3.110$ and $\mathbb{E}[\text{cost} \mid \text{do}(\pi = \text{passive})] = 16.106$; a Monte Carlo run with 20 000 fresh episodes per policy (`default_rng(7)`) gives 3.103 and 16.084 (Fig. 13.10).

PYTHON LAB · `s13_examples.py` · `ex_rl_counterfactual_replay()`

```
abduced noise      :  [ 0.514  0.821  0.573 -0.487 -0.696  0.034]
counterfactual     :  [ 2.     1.314  1.347  1.112 -0.042 -0.713 -0.252]
factual cost = 33.497;  counterfactual cost (same noise) = 6.280;  saving =
27.2
interventional (Monte Carlo, 20000 episodes each, seed=7) E[cost |
do(pi=feedback)] = 3.103; E[cost | do(pi=passive)] = 16.084
exact:  E[cost | do(pi=feedback)] = 3.110;  E[cost | do(pi=passive)] =
16.106
```

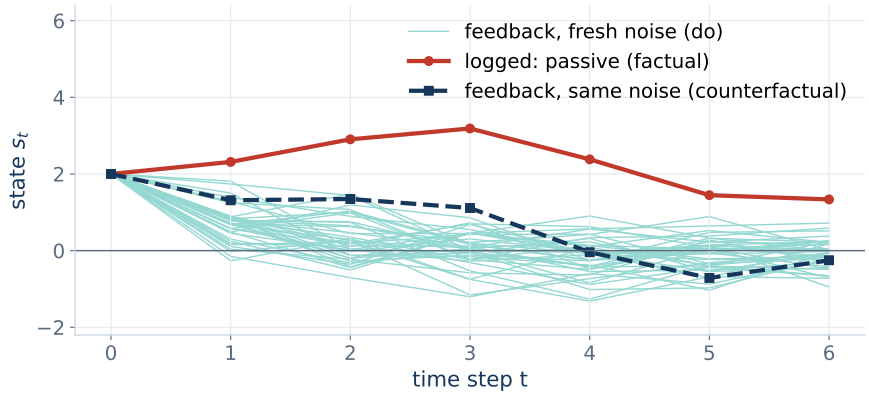


Figure 13.10. The logged passive episode of Worked Example 13.16 (factual), its counterfactual replay under the feedback policy $a_t = -s_t$ with the same abduced noise, and 40 episodes of the feedback policy under fresh noise (`default_rng(7)`), showing the interventional spread.

LINK TO COURSE SESSION 20 · Deep and Model-based RL

World models (Dyna, Dreamer-style latent models) are learned SCMs; "imagined" rollouts are samples from $P(\cdot \mid \text{do}(\pi))$. Their value depends on autonomy: if the learned f only captures correlations in the logged data (generated by an old policy), rollouts under a new policy can be badly wrong — the RL face of Worked Example 13.2.

13.12.3 Multi-agent AI: counterfactual credit assignment

In cooperative multi-agent RL all agents share one team reward, so each agent's gradient is swamped by its teammates' choices. Write s for the global state, $\mathbf{u} = (u_1, \dots, u_N)$ for the joint action, $\mathbf{u}^{-i}$ for the actions of all agents except i, τ^i for agent i's action–observation history, $\pi^i(u_i \mid \tau^i)$ for its policy, and $Q(s, \mathbf{u})$ for a centralized critic's estimate of the team return. COMA (Foerster et al. 2018) gives agent i the counterfactual advantage (the superscript i distinguishes it from the ability variable A of the wage SCM)

$$A^i(s, \mathbf{u}) = Q(s, \mathbf{u}) - \sum_{u_i'} \pi^i(u_i' \mid \tau^i)\, Q\big(s, (\mathbf{u}^{-i}, u_i')\big),$$

"how much better was my action than what I would have done on average, *holding my teammates' actions fixed*?" The baseline is a rung-3 quantity: it intervenes on agent i's action node while the other agents' realized actions are kept. Because the baseline b (the sum above) does not depend on u_i, it leaves the policy gradient unbiased: $\mathbb{E}_{u_i\sim\pi^i}[\nabla \log \pi^i(u_i)\, b] = \sum_{u_i} \pi^i(u_i) \frac{\nabla \pi^i(u_i)}{\pi^i(u_i)}\, b = b\,\nabla \sum_{u_i} \pi^i(u_i) = b\,\nabla 1 = 0$.

Worked Example 13.17 (COMA on a two-agent matrix)**.** Two agents with actions $\{a, b, c\}$ have centralized critic values $Q(u_1, u_2)$ (state fixed, illustrative):

$$Q = \begin{array}{c|ccc} & u_2 = a & u_2 = b & u_2 = c \\ \hline u_1 = a & 2 & 0 & 1 \\ u_1 = b & 4 & 3 & 0 \\ u_1 = c & 1 & 5 & 2 \end{array}$$

Policies over (a, b, c) are $\pi^1 = (0.5, 0.3, 0.2)$ and $\pi^2 = (0.4, 0.4, 0.2)$, and the joint action played was $(u_1, u_2) = (b, a)$ with $Q(b, a) = 4$. Agent 1's baseline varies its own action with agent 2 fixed at a (column a): $0.5 \cdot Q(a, a) + 0.3 \cdot Q(b, a) + 0.2 \cdot Q(c, a) = 0.5(2) + 0.3(4) + 0.2(1) = 1.0 + 1.2 + 0.2 = 2.4$, so $A^1 = 4 - 2.4 = 1.6$. Agent 2's baseline varies its action with agent 1 fixed at b (row b): $0.4(4) + 0.4(3) + 0.2(0) = 1.6 + 1.2 + 0 = 2.8$, so $A^2 = 4 - 2.8 = 1.2$. Both receive positive but different credit from the same team reward (Fig. 13.11).

PYTHON LAB · `s13_examples.py` · `ex_coma_baseline()`

```
agent 1:  baseline = 2.40, advantage = 1.60;  agent 2:  baseline = 2.80,
advantage = 1.20
```

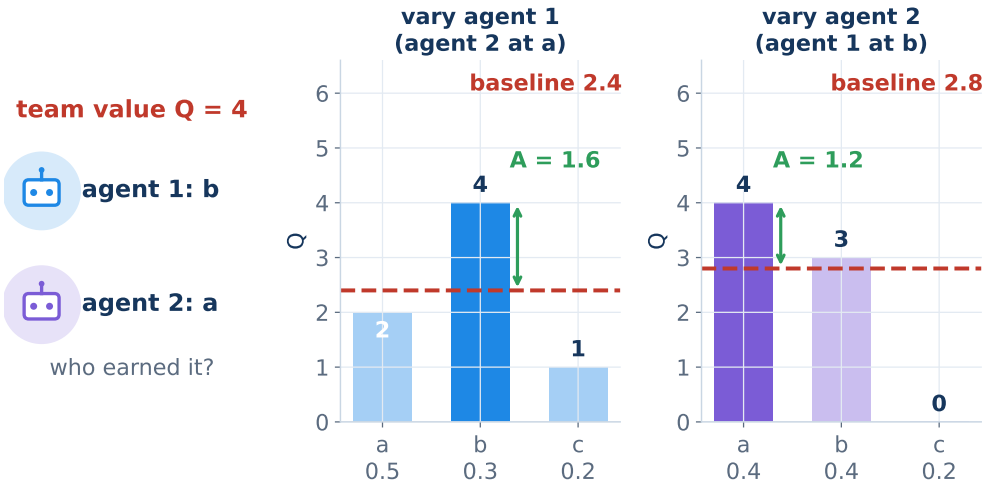


Figure 13.11. COMA's counterfactual baseline: each agent's action is compared with its own policy-average while the teammate's action is held at its realized value.

LINK TO COURSE SESSION 27 · Multi-Agent AI

Counterfactual reasoning appears throughout multi-agent AI: credit assignment (COMA, difference rewards $D_i = G(\mathbf{u}) - G(\mathbf{u}^{-i}, c_i)$, where G is the team reward and c_i a fixed default action), responsibility and blame in teams (PN for each agent), and mechanism design, where an agent's report is an intervention on a shared mechanism. Caveat: when one agent changes its policy the others adapt, so "holding the others fixed" is itself a modelling choice.

LINK TO COURSE SESSION 4 · Generative Models

VAE decoders, GAN generators and diffusion samplers map independent noise to data; all are push-forwards of noise (Proposition 13.3) but, without autonomy, encode only rung 1. Making the map *structural* (one mechanism per variable, independent noises) turns a generator into a simulator that answers do(·) and counterfactual queries — the core of counterfactual fairness (a decision should not change under do(protected attribute) for the same unit). Identifying the mechanisms (cf. Worked Example 13.3) needs extra structure such as non-Gaussianity or interventions. Likewise, a decoder $x = f_\theta(z)$ with independent latent z is a one-layer SCM with learned mechanisms, and counterfactual explanations of classifiers (Wachter, Mittelstadt and Russell 2018; course session 31) must propagate input changes through an SCM of the features, or they recommend impossible actions.

PART V · AI CONNECTIONS

13.13 Summary

Table 13.3. This session in the Advanced AI course.

Concept today	Where it reappears	How
SCM / autonomy	course 20, 23–24	world models; action schemas and the frame assumption
Graph surgery, do(·)	Session 14; course 17–19	identification; every RL action and policy is an intervention
Causal Markov, d-separation	Session 11; Session 14	PC algorithm tests implications; back-door criterion
Counterfactuals	Session 15; course 27, 31	ITE/CATE; COMA credit; counterfactual explanations
PN / PS / PNS	course 26, 31	agent self-reflection and blame; accountability
Push-forward of noise	course 4	VAEs, GANs, diffusion as (non-structural) generators

Table 13.4. Cheat-sheet of key formulas.

Object	Formula
SCM	$V_i := f_i(\mathrm{pa}_i, U_i)$, $P(\mathbf{U})$; Markovian if U_i independent
Causal Markov	$P(\mathbf{v}) = \prod_i P(v_i \mid \mathrm{pa}_i)$, $P(v_i \mid \mathrm{pa}_i) = P(f_i(\mathrm{pa}_i, U_i) = v_i)$
Global Markov	$\mathbf{X} \perp\!\!\!\perp_d \mathbf{Y} \mid \mathbf{Z}$ in $G \Rightarrow \mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z}$ in P
Truncated factorization	$P(\mathbf{v} \mid \mathrm{do}(x)) = \prod_{i:V_i \neq X} P(v_i \mid \mathrm{pa}_i)\big\vert_{X=x}$
Counterfactual	abduction $P(\mathbf{u} \mid e) \to$ action $M_{x'} \to$ prediction $Y_{x'}(\mathbf{u})$
PNS bounds	$\max(0, p_1 - p_0) \leq \mathrm{PNS} \leq \min(p_1, 1 - p_0)$; monotone: $p_1 - p_0$
Linear SEM	$\Sigma = (I - B)^{-1}\Omega(I - B)^{-\top}$; $\Sigma_{ij} = \sum_{\text{all treks}} \omega_{\text{top}} \prod b_e = \sum_{\text{simple treks}} \mathrm{Var}(V_{\text{top}}) \prod b_e$
Omitted-variable bias	$\boldsymbol{\beta} = \mathbf{b} + \Sigma_{XX}^{-1}\Sigma_{XA}\mathbf{c}$; scalar: $b + ac\sigma_A^2/(a^2\sigma_A^2 + \sigma_X^2)$
COMA advantage	$Q(s, \mathbf{u}) - \sum_{u_i'} \pi^i(u_i')Q(s, (\mathbf{u}^{-i}, u_i'))$

Table 13.5. Common pitfalls.

Pitfall	Correct view
Treating $:=$ as $=$	a structural assignment cannot be solved for its parents; regressions can
"The DAG fits the data, so it is causal"	Markov-equivalent SCMs fit equally well yet answer do$(\cdot)$ differently
Ignoring dependent noises	a hidden common cause breaks the causal Markov condition on the visible DAG
Equating do$(\cdot)$ with conditioning	conditioning keeps X's mechanism (and its confounding); surgery removes it
Regressing on "everything"	mediators, colliders and descendants of Y bias the coefficient of X

THINK ABOUT IT **The causal Markov condition needs independent noises. Which unmeasured factors would make the noises of "education" and "income" dependent, and how would you show that in the graph?**

THINK ABOUT IT **A world model in model-based RL is trained on logs from an old policy. Under which conditions is it structural (valid under the new policy) rather than merely predictive?**

THINK ABOUT IT **Why can randomized experiments identify PNS only up to an interval? What extra knowledge about an LLM agent's tool would collapse the interval?**

Further reading. Pearl (2009), *Causality*, chs. 1, 3, 7, 9; Pearl, Glymour and Jewell (2016), *Causal Inference in Statistics: A Primer*, chs. 2–4; Peters, Janzing and Schölkopf (2017), *Elements of Causal Inference*, chs. 3–6; Spirtes, Glymour and Scheines (2000), *Causation, Prediction, and*

Search; Hernán and Robins (2020), *Causal Inference: What If*, ch. 6; Koller and Friedman (2009), ch. 21; Foerster et al. (2018), "Counterfactual multi-agent policy gradients", AAAI.

APPENDIX

13.14 Appendix: General Confounding-Bias Formula for Linear SCMs

For the general linear SCM $X = aA + U_X$, $Y = bX + cA + U_Y$ (all noises independent, unit variance, A unit variance), we have $\mathrm{Var}(X) = a^2\mathrm{Var}(A) + \mathrm{Var}(U_X) = a^2 + 1$, $\mathrm{Cov}(X, A) = a$ and $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X,\, bX + cA + U_Y) = b\mathrm{Var}(X) + c\mathrm{Cov}(X, A) = b(a^2 + 1) + ac$, so the observational regression slope of Y on X is $\dfrac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)} = \dfrac{b(a^2+1)+ac}{a^2+1} = b + \dfrac{ac}{a^2+1}$, while the true causal effect is exactly b (from $\mathrm{do}(X = x)$: $\mathbb{E}[Y \mid \mathrm{do}(x)] = bx + c\mathbb{E}[A] = bx$ when $\mathbb{E}[A] = 0$). The bias term $\dfrac{ac}{a^2+1}$ vanishes exactly when $a = 0$ (no confounder $\to X$ edge) or $c = 0$ (no confounder $\to Y$ edge) — confirming that confounding bias requires *both* arrows of the fork to be present, consistent with the graphical picture throughout this session. Plugging in $a = 2, b = 3, c = 5$ recovers the bias in Worked Example 13.10, $\frac{2\cdot 5}{4+1} = \frac{10}{5} = 2$, i.e., observational slope $3 + 2 = 5$, matching both the hand calculation and the simulation. In trek language (Theorem 13.5) the bias is the single trek $X \leftarrow A \rightarrow Y$ divided by $\mathrm{Var}(X)$, and Proposition 13.4 gives the multivariate version.

CONNECTIONS AND TAKEAWAYS

We now have a mathematical object — the SCM — in which "the effect of an intervention" and even "what would have happened to this specific unit" are well-defined quantities, computed by the truncated factorization formula (Theorem 13.3) directly from the same graphical and computational machinery built in Sessions 6–7. The causal Markov condition (Theorem 13.1) tells us exactly which observable constraints a causal model imposes, and d-separation reads them off the graph. This resolves Session 12's conceptual gap in principle. But notice the catch in every worked example above: we always assumed we had the *full structural equations*, including the confounder A and its exact numerical role. In practice we almost never observe every relevant confounder, let alone know its functional form. **Session 14** asks the sharper, practitioner-relevant question: if all we are given is an observational dataset plus a causal *graph* (no equations, and possibly some variables, like A, unmeasured, i.e. bidirected edges) can we still compute $P(y \mid \mathrm{do}(x))$ using only the pieces we can actually observe? The answer will be: sometimes yes, via a checkable graphical criterion — the back-door and front-door criteria and the do-calculus — and sometimes provably no.